MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

LATCH, DIGITAL, KEY CODE GBL

INACTIVE FOR NEW DESIGN
AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic latch module.

1.2 Classification. This specification covers one class of modules.

Class 1 as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.11 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GBL.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/44-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

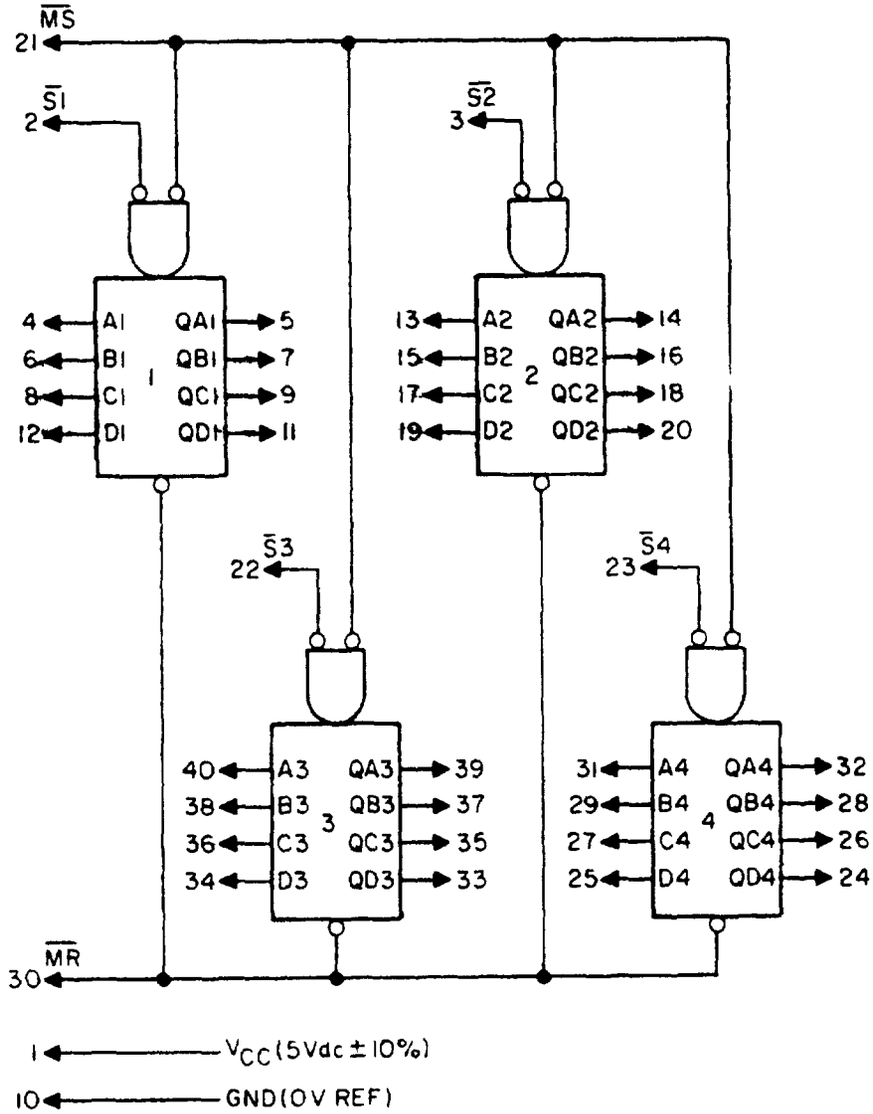
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES.
1. A, B, C, D = Latch inputs.
2. $\overline{S}$ = ANDed strobe input (active low).
3. $\overline{MS}$ = Master strobe (active low).
4. $\overline{MR}$ = Master reset (active low).
5. QA, QB, QC, QD = Latch outputs.

* FIGURE 1. GBL module function diagram, latch.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A16	V_{OUT} (low loaded)	III	S	100	600	100	600	100	600	100	600	mV
B1-B20	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	mA
B21, B22	Input breakdown current	IV	S	0	4	0	4	0	4	0	4	mA
C	Isolation	V	S	10	-	1	-	10	-	1	-	MΩ
D1-D22	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1 5	V
E1-E16	Input current, high level	VII	X	0	30	0	30	0	30	0	30	µA
E17-E20	Input current, high level	VII	X	0	20	0	20	0	20	0	20	µA
E21, E22	Input current, high level	VII	X	0	80	0	80	0	80	0	80	µA
F1-F16	Input current, low level	VII	X	10	640	10	640	10	640	10	640	µA
F17-F20	Input current, low level	VII	X	10	430	10	430	10	430	10	430	µA
F21, F22	Input current, low level	VII	X	0.04	1.72	0.04	1.72	0.04	1.72	0.04	1.72	mA
G1-G16	Output short circuit current	VII	X	2.5	25.0	2.5	25.0	2.5	25.0	2.5	25.0	mA
H1-H48	Output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J64	Output voltage, low level	VIII	X	0	300	0	300	0	300	0	300	mV
K	Power supply current (all outputs low level)	IX	X	10	66	10	66	10	66	10	66	mA
L1-L16	Transition time (t_{TLH}), C_L = 50 pF	X	S	5	40	5	40	5	40	5	40	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1-M16	Transition time (t_{TLH}), C_L = 150 pF	X	Q	5	60	5	60	5	60	5	60	ns
N1-N16	Transition time (t_{THL}), C_L = 50 pF	X	S	2	30	2	30	2	30	2	30	ns
P1-P16	Transition time (t_{THL}), C_L = 150 pF	X	Q	2	40	2	40	2	40	2	40	ns
Q1-Q16	Propagation delay time (t_{PLH}), C_L = 50 pF	XI	S	10	60	10	60	10	65	10	65	ns
R1-R16	Propagation delay time (t_{PLH}), C_L = 150 pF	XI	Q	10	75	10	75	20	80	10	80	ns
S1-S16	Propagation delay time (t_{PHL}), C_L = 50 pF	XI	S	10	65	10	65	10	70	10	70	ns
T1-T16	Propagation delay time (t_{PHL}), C_L = 150 pF	XI	Q	10	80	10	80	10	85	10	85	ns
U1-U16	Propagation delay time (t_{PLH}), C_L = 50 pF	XI	S	10	80	10	80	10	85	10	85	ns
V1-V16	Propagation delay time (t_{PLH}), C_L = 150 pF	XI	Q	10	95	10	95	10	100	10	100	ns
W1-W16	Propagation delay time (t_{PHL}), C_L = 50 pF	XI	S	10	65	10	65	10	70	10	70	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
X1-X16	Propagation delay time (t_{PHL}), C_L = 150 pF	XI	Q	10	80	10	80	10	85	10	85	ns
Y1-Y16	Propagation delay time (t_{PLH}), C_L = 50 pF	XI	S	10	80	10	80	10	85	10	85	ns
Z1-Z16	Propagation delay time (t_{PLH}), C_L = 150 pF	XI	Q	10	95	10	95	10	100	10	100	ns
AA1-AA16	Propagation delay time (t_{PHL}), C_L = 50 pF	XI	S	10	65	10	65	10	70	10	70	ns
BB1-BB16	Propagation delay time (t_{PHL}), C_L = 150 pF	XI	Q	10	80	10	80	10	85	10	85	ns
CC1-CC16	Setup time check	XII	S	175	205	175	205	175	205	175	205	ns
DD1-DD16	Hold time check	XII	S	185	215	185	215	185	215	185	215	ns
EE1-EE16	Setup time check	XII	S	175	205	175	205	175	205	175	205	ns
FF1-FF16	Hold time check	XII	S	185	215	185	215	185	215	185	215	ns
GG1-GG16	Master reset delay time, C_L = 50 pF	XIII	S	10	65	10	65	10	70	10	70	ns
GG17-GG32	Master reset delay time, C_L = 50 pF, PWC	XIII	S	400	600	400	600	400	600	400	600	ns
HH1-HH16	Master reset delay time, C_L = 150 pF	XIII	Q	10	80	10	80	10	85	10	85	ns
HH17-HH32	Master reset delay time, C_L = 150 pF, PWC	XIII	Q	400	600	400	600	400	600	400	600	ns
JJ1-JJ16	Master reset recovery time check	XIV	S	180	220	180	220	180	220	180	220	ns
KK	Power supply filter capacitance	XV	S	1 6	3 0	1.6	3.0	1.6	3.0	1.6	3 .0	µF
LL1-LL16	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
MM1-MM16	Output crosstalk, low level	XVI	Q	0	750	0	750	0	750	0	750	mV
MM17-MM32	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
NN	Power supply transient amplitude	XVII	Q	0	300	0	300	0	300	0	300	mV
PP1-PP20	Input capacitance	XVIII	Q	2	15	2	15	2	15	2	15	pF
PP21, PP22	Input capacitance	XVIII	Q	8	45	8	45	8	45	8	45	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through PP shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through PP in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through NN shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition KK, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests D through NN shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through PP shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross-reference of classification. Part number M28787/44-1 supersedes part number M28787/44. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage	5 Vdc ±10 percent
Maximum power dissipation (All outputs at low level and not loaded and all inputs and strobe disabled)	363 mW

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Input voltage		
Low level		-0.5 V minimum, 0.7 V maximum
High level		2.0 V minimum, V_{CC} maximum
Input current (A, B, C, and D)		
Low level	(V_{IL} = 0.0 V)	-0.64 mA maximum
	(V_{IL} = 0.3 V)	-0.6 mA maximum
High level	(V_{IH} = 2.4 V)	30 µA maximum
	(V_{IH} = 5.5 V	1 mA maximum
Input current $\overline{S}$		
Low level	(V_{IL} = 0.0 V)	-0.43 mA maximum
	(V_{IL} = 0.3 V)	-0.4 mA maximum
High level	(V_{IH} = 2.4 V)	20 µA maximum
	(V_{IH} = 5.5 V)	1 mA maximum
Input current ($\overline{MS}$ and $\overline{MR}$		
Low level	(V_{IL} = 0.0 V)	-1.72 mA maximum
	(V_{IL} = 0.3 V)	-1.6 mA maximum
High level	(V_{IH} = 2.4 V)	80 µA maximum
	(V_{IH} = 5.5 V)	4 mA maximum
Output voltage		
Low level		0.0 V minimum, 0.3 V maximum
High level		2.4 V minimum, V_{CC} maximum
Output drive current maximum capability		
Low level	(V_{OL} = 0.3 V)	3.6 mA
High level	(V_{OH} = 2.4 V)	-0.36 mA
Pulse width ($\overline{S}$)		45 ns minimum
Pulse width ($\overline{MR}$)		40 ns minimum
Recovery time ($\overline{MR}$)		30 ns minimum
Propagation delay time		
$\overline{S}$ to QA-QD		
t_{PHL}	(C_L = 50 pF)	70 ns maximum
	(C_L = 150 pF)	85 ns maximum
t_{PLH}	(C_L = 50 pF)	85 ns maximum
	(C_L = 150 pF)	100 ns maximum

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Propagation delay time		
A-D to QA-QD		
t_{PHL}	(C_L = 50 pF)	70 ns maximum
	(C_L = 150 pF)	85 ns maximum
t_{PLH}	(C_L = 50 pF	65 ns maximum
	(C_L = 150 pF	80 ns maximum
Delay time (MR)		
C_L = 50 pF		70 ns maximum
C_L =150 pF		85 ns maximum
Setup and hold time (A-D to $\overline{S}$)		
SETUP (H)		25 ns minimum
HOLD (H)		15 ns maximum
Setup (L)		35 ns minimum
HOLD (L)		15 ns maximum

* 6.5 Module function. This module contains four 4-input low power storage latches with four corresponding true outputs (QA-QD). Each 4-bit latch has one strobe (S) input ANDed with a master strobe (MS) input. All four latches are controlled by one master strobe (MS) input (pin 21) and one master reset (MR) input (pin 30).

6.5.1 Data entry. Data can be entered into the latch when both the individual strobe ($\overline{S}$) and the master strobe ($\overline{MS}$) inputs are low. As long as this logic condition exists, the output of the latch will follow the input. If either of the strobe inputs ($\overline{MS}$ or $\overline{S}$) goes high, the data present in the latch at the time is held in the latch and is no longer affected by the data input.

6.5.2 Master reset override. The master reset ($\overline{MR}$) overrides all other input conditions and forces the outputs of all the latches low when a low signal is applied to the master reset ($\overline{MR}$) input (pin 30).

6.5.3 Input to output relationship. Each latch has the input to output relationship of table II and figure 2 for positive logic.

6.5.4 Parallel outputs (wired OR). Paralleling of outputs is not permissible.

6.5.5 Crosstalk relationships. The module has the following crosstalk relationships.

6.5.5.1 Module crosstalk. When connected in the worst case mode of operation (all outputs but one switching simultaneously from a high level to a low level), the GBL module could exhibit substantial crosstalk. When the quiet line is a low level, there may be as much as 750 mV positive going spike coincident with the negative edge of the switching outputs.

6.5.5.2 Circuit crosstalk. The crosstalk on any output pin is nearly directly proportional to the number of switching outputs. For every switching output in circuit 1, the noise induced on a low level quiet output in circuit 1 is as much as 110 mV. For every switching output outside circuit 1 the noise induced on a low level quiet output in circuit 1 is as much as 35 mV. The preceding relationships also hold for circuits 2-4.

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6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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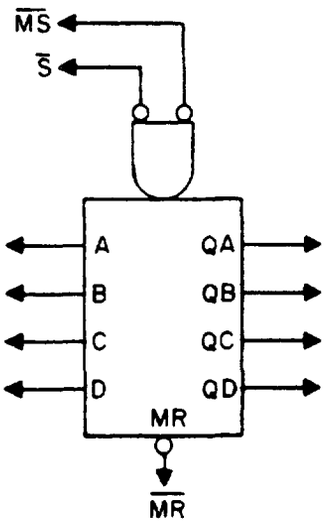
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TABLE II. Input to output relationship. 1/ - 5/

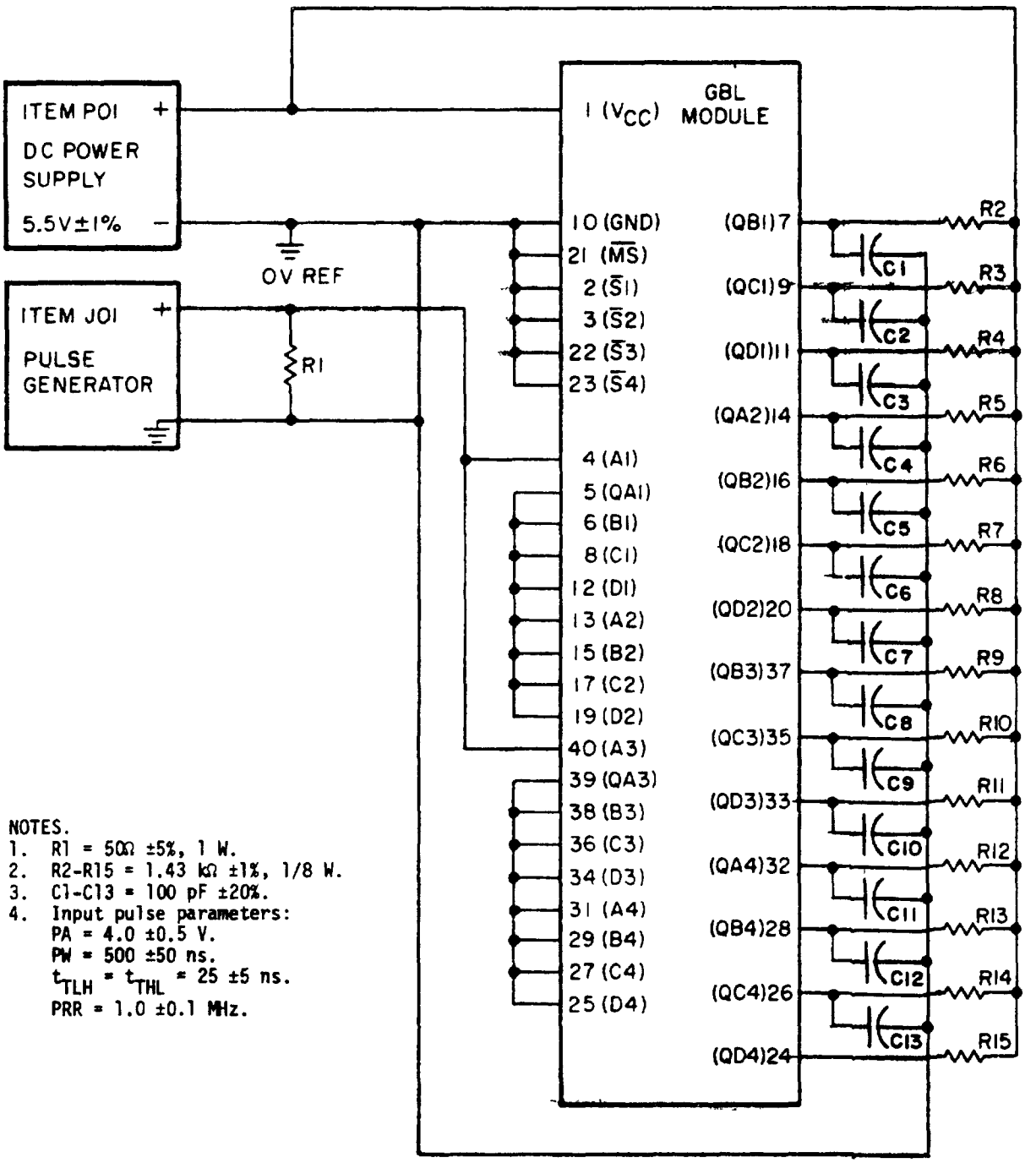
$\overline{MR}$	$\overline{MS}$	$\overline{S}$	Data	Qn	Operation
H	L	L	L	L	Data entry
H	L	L	H	H	Data entry
H	L	H	X	Qn-1	Hold
H	H	L	X	Qn-1	Hold
H	H	H	X	Qn-1	Hold
L	X	X	X	L	Reset

1/ X = Don't care.
2/ L = Low-voltage level.
3/ H = High-voltage level.
4/ Qn-1 = Previous output state.
5/ Qn = Present output state.



* FIGURE 2. Input to output relationship.

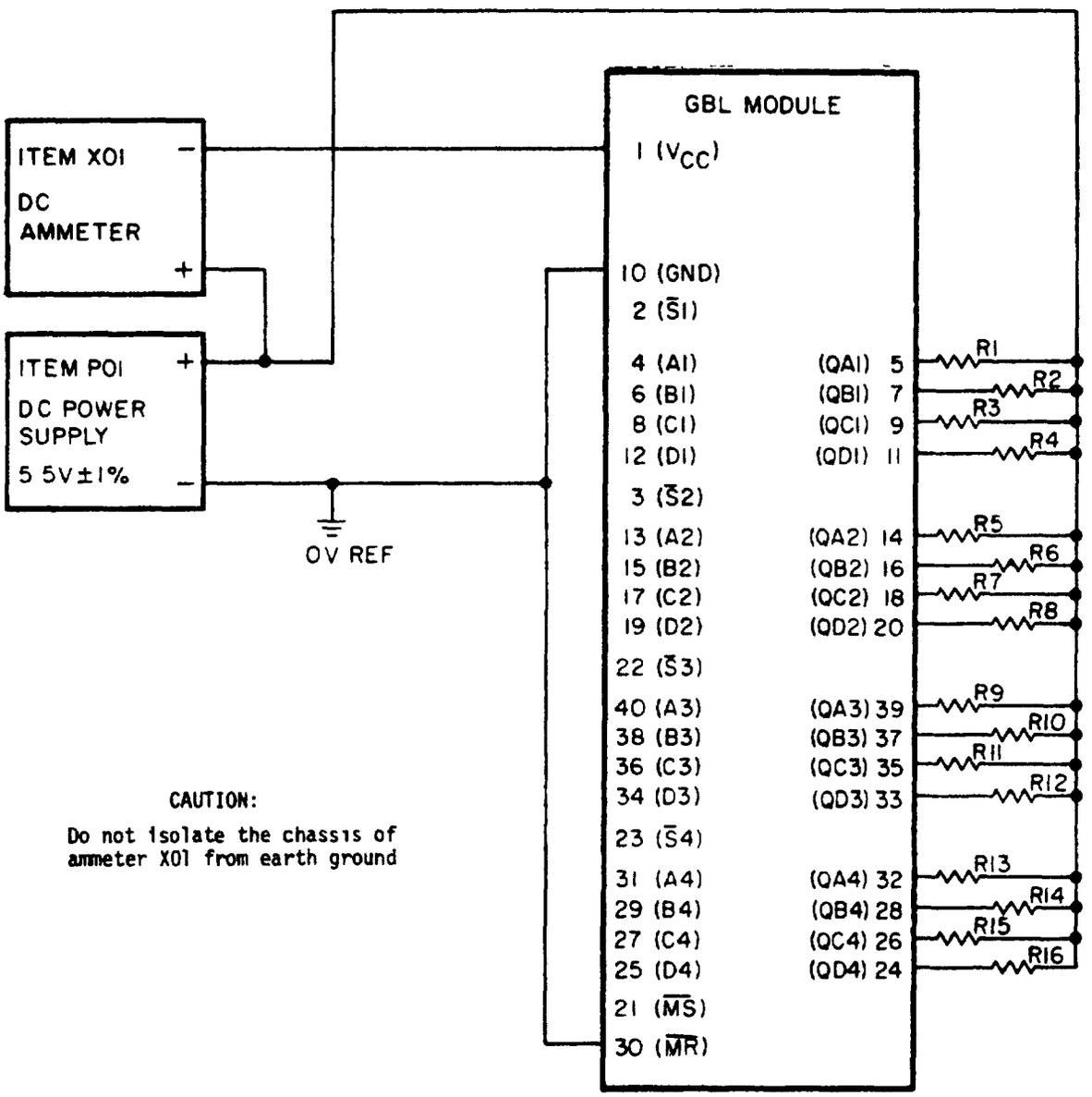
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NOTES.
1. R1 = 50Ω ±5%, 1 W.
2. R2-R15 = 1.43 kΩ ±1%, 1/8 W.
3. C1-C13 = 100 pF ±20%.
4. Input pulse parameters:
   PA = 4.0 ±0.5 V.
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   PRR = 1.0 ±0.1 MHz.

FIGURE 3. Operating temperature and life test circuit.

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NOTE: R1-R16 = 1.43 kΩ ±1%, 1/8 W.

FIGURE 4. Component temperature test circuit.

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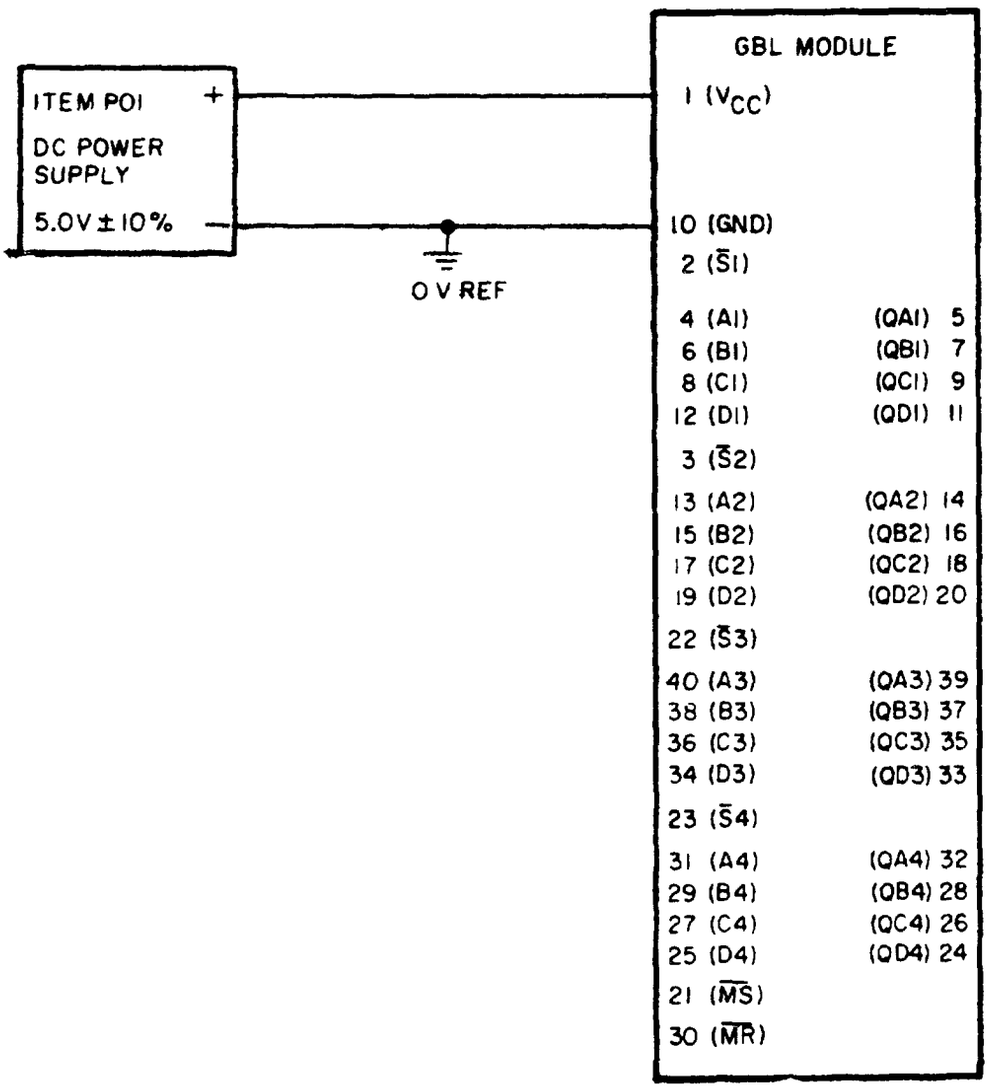


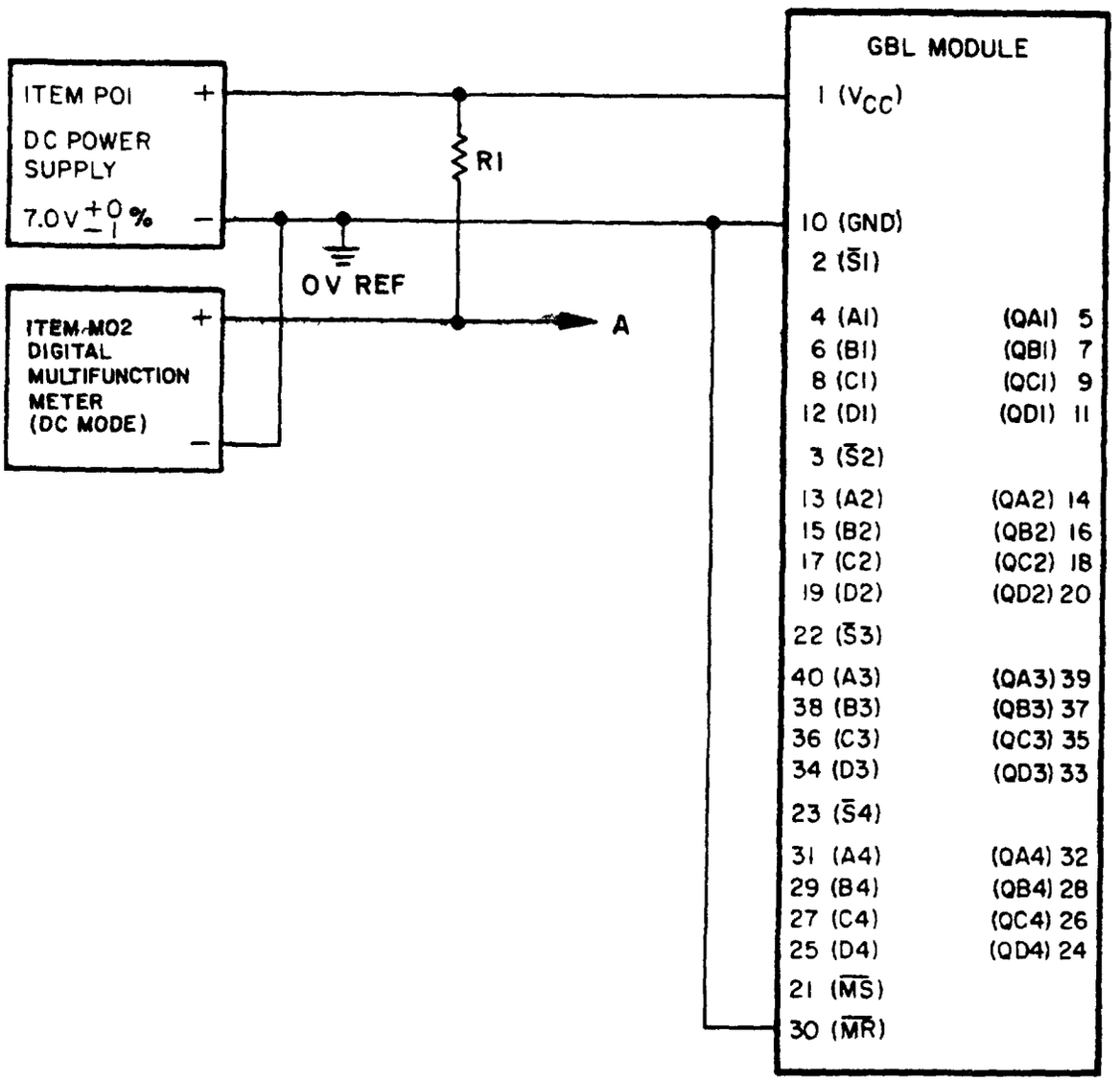
FIGURE 5. Vibration test circuit.

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* TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	QA1 V_{OUT} (low loaded)	5	100	600	mV	6
A2	QB1	7	↓	↓	↓	↓
A3	QC1	9	↓	↓	↓	↓
A4	QD1	11	↓	↓	↓	↓
A5	QA2	14	↓	↓	↓	↓
A6	QB2	16	↓	↓	↓	↓
A7	QC2	18	↓	↓	↓	↓
A8	QD2	20	↓	↓	↓	↓
A9	QA3	39	↓	↓	↓	↓
A10	QB3	37	↓	↓	↓	↓
A11	QC3	35	↓	↓	↓	↓
A12	QD3	33	↓	↓	↓	↓
A13	QA4	32	↓	↓	↓	↓
A14	QB4	28	↓	↓	↓	↓
A15	QC4	26	↓	↓	↓	↓
A16	QD4	24	↓	↓	↓	↓

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NOTE: R1 = 1.43 kΩ ±1%, 1/8 W.

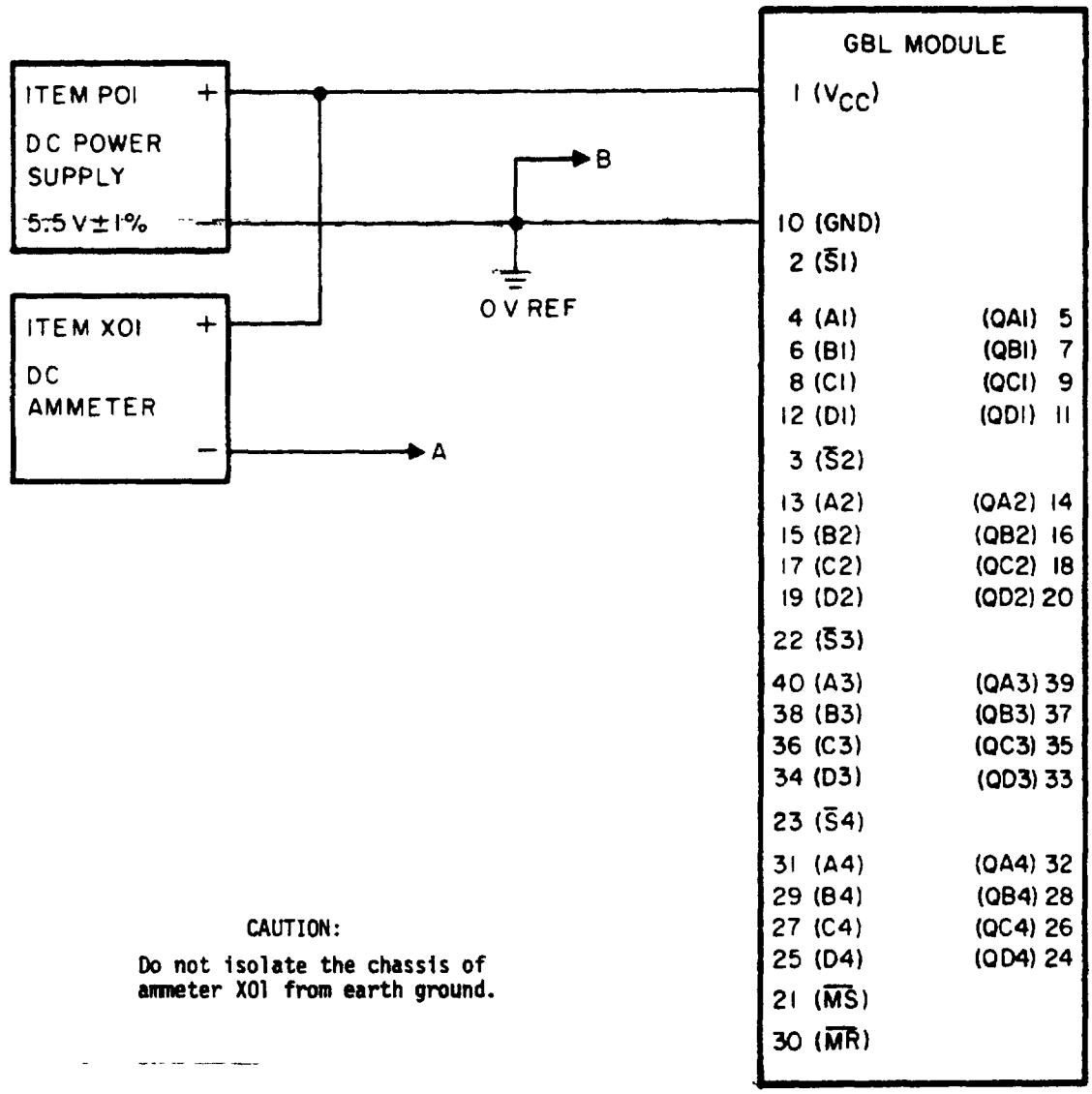
* FIGURE 6. V_{OUT} (low-loaded)' test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	A1 Input breakdown current	4	30	0	1	mA	7
B2	B1	6					
B3	C1	8					
B4	D1	12					
B5	A2	13					
B6	B2	15					
B7	C2	17					
B8	D2	19					
B9	A3	40					
B10	B3	38					
B11	C3	36					
B12	D3	34					
B13	A4	31					
B14	B4	29					
B15	C4	27					
B16	D4	25					
B17	$\overline{S1}$	2	21				
B18	$\overline{S2}$	3					
B19	$\overline{S3}$	22					
B20	$\overline{S4}$	23					
B21	MS	21	2, 3, 22, 23		4		
B22	MR	30	Open		4		

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

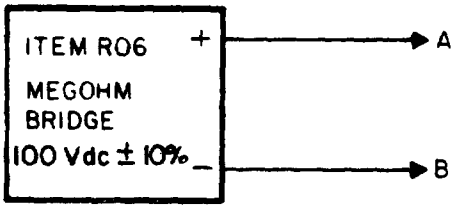
FIGURE 7. Input breakdown current test circuit for table IV.

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TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Isolation	All pins	Frame	10	∞	MΩ	8

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WARNING.

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

GBL MODULE	
1 (V_{CC})	
10 (GND)	
2 ($\overline{S}1$)	
4 (A1)	(QA1) 5
6 (B1)	(QB1) 7
8 (C1)	(QC1) 9
12 (D1)	(QD1) 11
3 ($\overline{S}2$)	
13 (A2)	(QA2) 14
15 (B2)	(QB2) 16
17 (C2)	(QC2) 18
19 (D2)	(QD2) 20
22 ($\overline{S}3$)	
40 (A3)	(QA3) 39
38 (B3)	(QB3) 37
36 (C3)	(QC3) 35
34 (D3)	(QD3) 33
23 ($\overline{S}4$)	
31 (A4)	(QA4) 32
29 (B4)	(QB4) 28
27 (C4)	(QC4) 26
25 (D4)	(QD4) 24
21 ($\overline{MS}$)	
30 ($\overline{MR}$)	

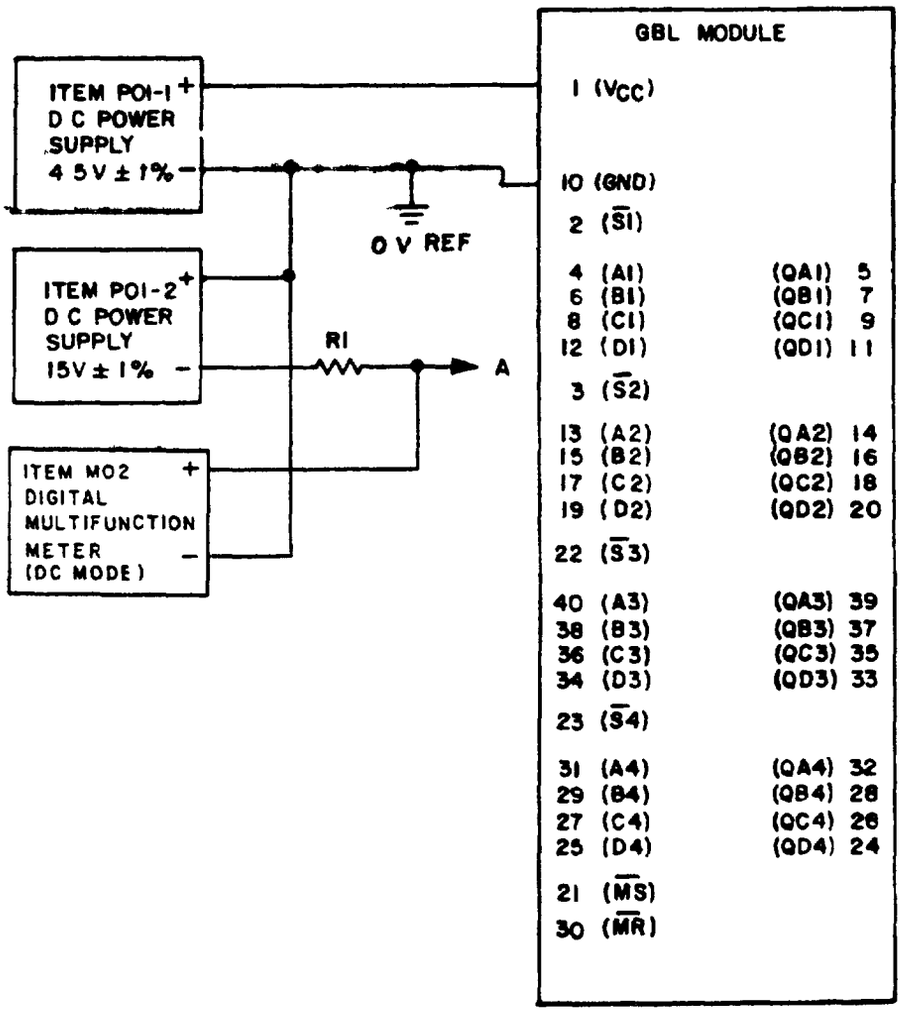
* FIGURE 8. Isolation test circuit for table V.

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TABLE VI. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	A1 Input clamp diode voltage	4	-0.4	-1.5	V	9
D2	B1	6				
D3	C1	8				
D4	D1	12				
D5	A2	13				
D6	B2	15				
D7	C2	17				
D8	D2	19				
D9	A3	40				
D10	B3	38				
D11	C3	36				
D12	D3	34				
D13	A4	31				
D14	B4	29				
D15	C4	27				
D16	D4	25				
D17	$\overline{S1}$	2				
D18	$\overline{S2}$	3				
D19	$\overline{S3}$	22				
D20	$\overline{S4}$	23				
D21	$\overline{MS}$	21				
D22	$\overline{MR}$	30				

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NOTE: RI = 1.13 kΩ ±1%, 1/8 W.

* FIGURE 9. Input clamp diode voltage test circuit for table VI.

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* TABLE VII. Input current (high-level and low-level) and output short circuit current test.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	A1	Input current, high level	4	Open	30	0	30	µA	10
E2	B1		6						
E3	C1		8						
E4	D1		12						
E5	A2		13						
E6	B2		15						
E7	C2		17						
E8	D2		19						
E9	A3		40						
E10	B3		38						
E11	C3		36						
E12	D3		34						
E13	A4		31						
E14	B4		29						
E15	C4		27						
E16	D4		25						
E17	$\overline{S1}$		2		21		20		
E18	$\overline{S2}$		3						
E19	$\overline{S3}$		22						
E20	$\overline{S4}$		23						
E21	$\overline{MS}$		21		2, 3, 22, 23		80		
E22	$\overline{MR}$		30		Open		80		

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* TABLE VII. Input current (high-level and low-level) and output short circuit current test - Continued.

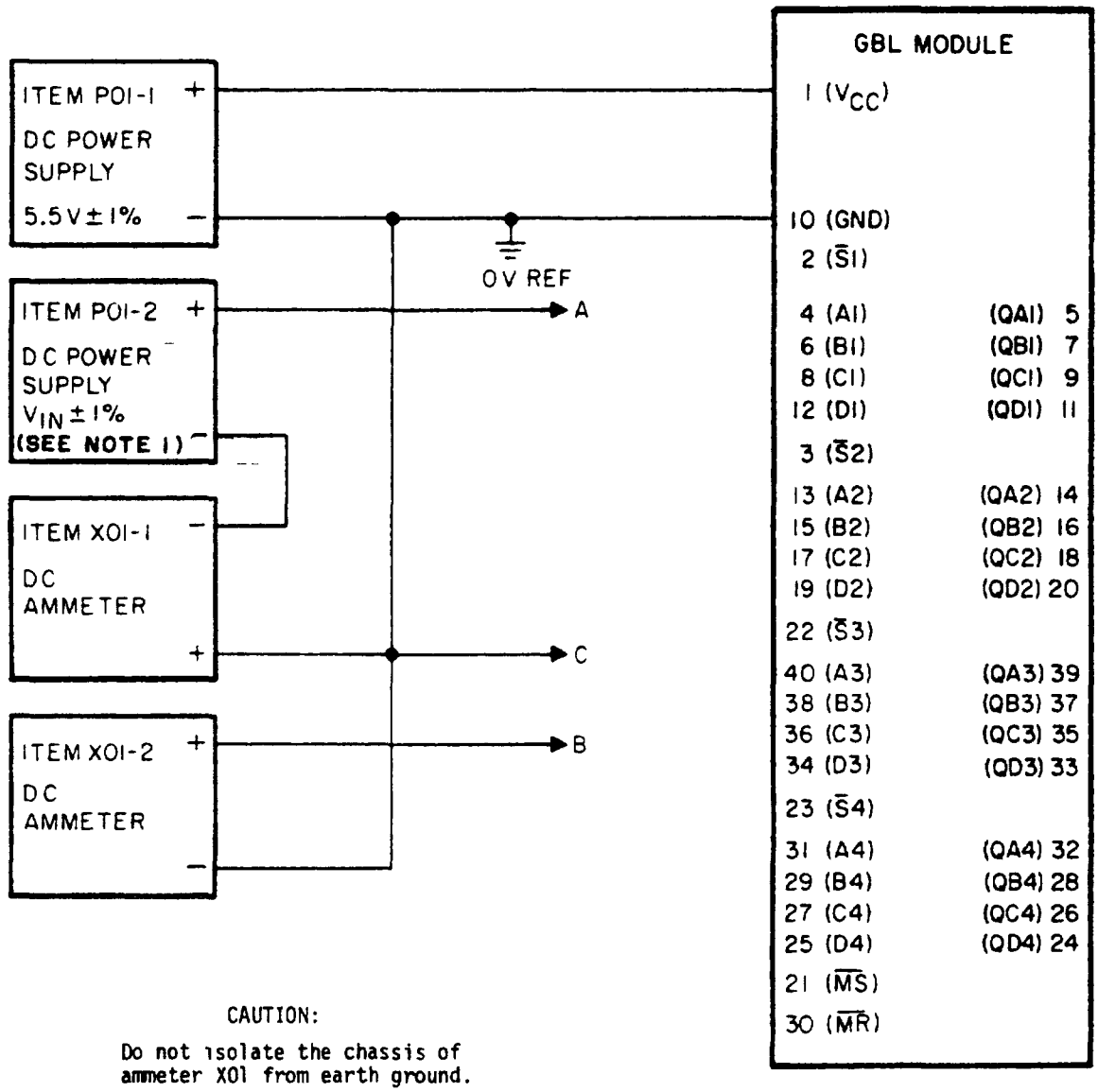
Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F1	A1	Input current, low level	Open	4	2, 21	10	640	µA	10
F2	B1			6					
F3	C1			8					
F4	D1			12					
F5	A2			13	3, 21				
F6	B2			15					
F7	C2			17					
F8	D2			19					
F9	A3			40	22, 21				
F10	B3			38					
F11	C3			36					
F12	D3			34					
F13	A4			31	23, 21				
F14	B4			29					
F15	C4			27					
F16	D4			25					
F17	$\overline{S1}$		21	2	Open		430		
F18	$\overline{S2}$			3					
F19	$\overline{S3}$			22					
F20	$\overline{S4}$			23					
F21	$\overline{MS}$		2, 3, 22, 23	21		0.04	1.72	mA	
F22	$\overline{MR}$		Open	30		0.04	1.72	mA	

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* TABLE VII. Input current (high-level and low-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	QA1	Outut short circuit current	4	5	2, 21	2.5	25.0	mA	10
G2	QB1		6	7					
G3	QC1		8	9					
G4	QD1		12	11					
G5	QA2		13	14	3, 21				
G6	QB2		15	16					
G7	QC2		17	18					
G8	QD2		19	20					
G9	QA3		40	39	22, 21				
G10	QB3		38	37					
G11	QC3		36	35					
G12	QD3		34	33					
G13	QA4		31	32	23, 21				
G14	QB4		29	28					
G15	QC4		27	26					
G16	QD4		25	24					

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Adjust item P01-2 voltage as follows:
   V_{IN} = 2.4 V ±1% for test conditions E1-E22.
   V_{IN} = 4.5 V ±1% for test conditions F1-F22 and G1-G16.
2. Use item X01-1 for test conditions E1-E22 and item X01-2 for test conditions F1-F22 and G1-G16.
3. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
4. With point A disconnected, item X01-1 should indicate 0.0 current on the 1.0 µA range.
5. When performing the output short circuit current test, only one output shall be shorted at a time, and the measurement taken within five seconds.

* FIGURE 10. Input current (high-level and low-level) and output short circuit current test circuit for table VII.

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TABLE VIII. Output voltage (high-level and low-level) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H1	QA1 Output voltage, high level	Open	4	5	Open	Open	2,21	2.4	4.5	V	11
H2	QB1		6	7							
H3	QC1		8	9							
H4	QD1		12	11							
H5	QA2		13	14			3,21				
H6	QB2		15	16							
H7	QC2		17	18							
H8	QD2		19	20							
H9	QA3		40	39			22, 21				
H10	QB3		38	37							
H11	QC3		36	35							
H12	QD3		34	33							
H13	QA4		31	32							
H14	QB4		29	28							
H15	QC4		27	26							
H16	QD4		25	24							
H17	QA1	2, 21	Open	5	30	4	Open				
H18	QB1			7		6					
H19	QC1			9		8					
H20	QD1			11		12					
H21	QA2	3, 21		14		13					
H22	QB2			16		15					
H23	QC2			18		17					
H24	QD2			20		19					

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TABLE VIII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H25	QA3 Output voltage, high level	22, 21	Open	39	30	40	Open	2.4	4.5	V	11
H26	QB3			37		38					
H27	QC3			35		36					
H28	QD3			33		34					
H29	QA4	23, 21		32		31					
H30	QB4			28		29					
H31	QC4			26		27					
H32	QD4			24		25					
H33	QA1	Open	30	5	Open	4	2, 21				
H34	QB1			7		6					
H35	QC1			9		8					
H36	QD1			11		12					
H37	QA2			14		13					
H38	QB2			16		15					
H39	QC2			18		17					
H40	QD2			20		19					
H41	QA3			39		40	22, 21				
H42	QB3			37		38					
H43	QC3			35		36					
H44	QD3			33		34					
H45	QA4			32		31	23, 21				
H46	QB4			28		29					
H47	QC4			26		27					
H48	QD4			24		25					

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* TABLE VIII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
J1	QA1	Output voltage, low level	4	Open	5	Open	Open	2, 21	0	300	mV	11
J2	QB1		6		7							
J3	QC1		8		9							
J4	QD1		12		11							
J5	QA2		13		14			3,21				
J6	QB2		15		16							
J7	QC2		17		18							
J8	QD2		19		20							
J9	QA3		40		39			22, 21				
J10	QB3		38		37							
J11	QC3		36		35							
J12	QD3		34		33							
J13	QA4		31		32			23, 21				
J14	QB4		29		28							
J15	QC4		27		26							
J16	QD4		25		24							
J17	QA1		Open	2	5	30	4	21				
J18	QB1				7		6					
J19	QC1				9		8					
J20	QD1				11		12					
J21	QA2			3	14		13					
J22	QB2				16		15					
J23	QC2				18		17					
J24	QD2				20		19					

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* TABLE VIII. Output voltage (high-level and low-level) test - Continued.

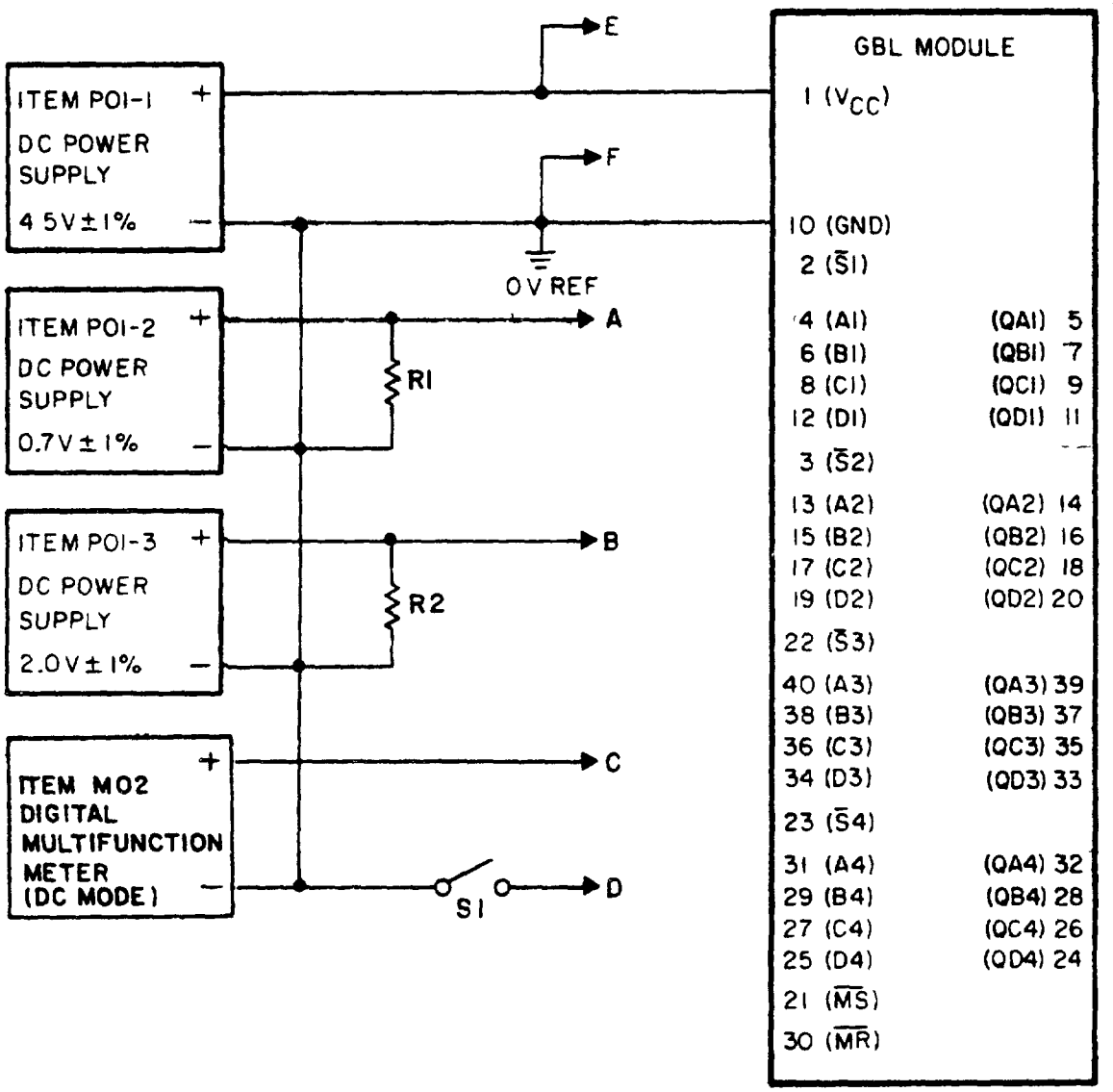
Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
J25	QA3 Output voltage, low level	Open	22	39	30	40	21	0	300	mV	11
J26	QB3			37		38					
J27	QC3			35		36					
J28	QD3			33		34					
J29	QA4		23	32		31					
J30	QB4			28		29					
J31	QC4			26		27					
J32	QD4			24		25					
J33	QA1		21	5		4	2				
J34	QB1			7		6					
J35	QC1			9		8					
J36	QD1			11		12					
J37	QA2			14		13	3				
J38	QB2			16		15					
J39	QC2			18		17					
J40	QD2			20		19					
J41	QA3			39		40	22				
J42	QB3			37		38					
J43	QC3			35		36					
J44	QD3			33		34					
J45	QA4			32		31	23				
J46	QB4			28		29					
J47	QC4			26		27					
J48	QD4			24		25					

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* TABLE VIII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
J49	QA1 Output voltage, low level	30	Open	5	Open	4	2, 21	0	300	mV	11
J50	QB1			7		6					
J51	QC1			9		8					
J52	QD1			11		12					
J53	QA2			14		13	3, 21				
J54	QB2			16		15					
J55	QC2			18		17					
J56	QD2			20		19					
J57	QA3			39		40	22, 21				
J58	QB3			37		38					
J59	QC3			35		36					
J60	QD3			33		34					
J61	QA4			32		31	23, 21				
J62	QB4			28		29					
J63	QC4			26		27					
J64	QD4			24		25					

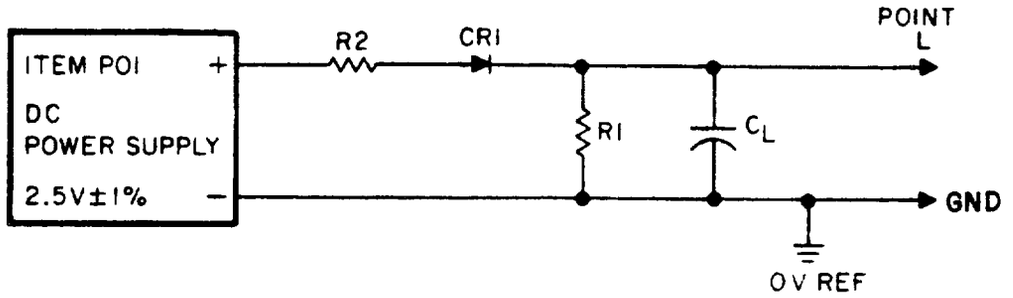
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. S1 = SPST switch.
3. After making all circuit connections, close switch S1 for a minimum of 100 ns; then open and allow a minimum of 50 ns before recording measurement.
4. Load all outputs with a load as specified on figure 12.

* FIGURE 11. Output voltage (high-level and low-level) test circuit for table VIII.

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NOTES:

1. Load circuit 1 for test conditions H, J, L, N, Q, S, U, W, Y, AA, CC, DD, EE, FF, GG, JJ, LL and MM.

   a. CR1 = 1N4150 or equivalent.
   b. C_L = 50 ±5 pF (including probe and parasitic capacitance).
c c. R1 = 6.65 kΩ ±1%, 1/8 W.
* d. R2 = Select value for a current flow of 3.6 mA ±5% out of point L with point L held at 0.3 V.
   e. Capacitor C_L is not required for dc electrical test conditions H or J.
   f. During output switching, item POI shall have a maximum overshoot of 500 mV peak.

2. Load circuit 2 for test conditions M, P, R, T, V, X, Z, BB and HH

   a. CR1 = 1N4150 or equivalent.
   b. C_L = 150 ±5 pF (including probe and parasitic capacitance).
   c. R1 = 6.65 kΩ ±1%, 1/8 W (current flow of 0.36 mA when point L is held at 2.4 V).
* d. R2 = Select value for a current flow of 3.6 mA ±5% out of point L with point L held at 0.3 V.
   e. During output switching, item POI shall have a maximum overshoot of 500 mV peak.

FIGURE 12. GBL logic load circuit.

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TABLE IX. Power supply current (all outputs low level) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K	Power supply current, all outputs low level	30	10	66	mA	13

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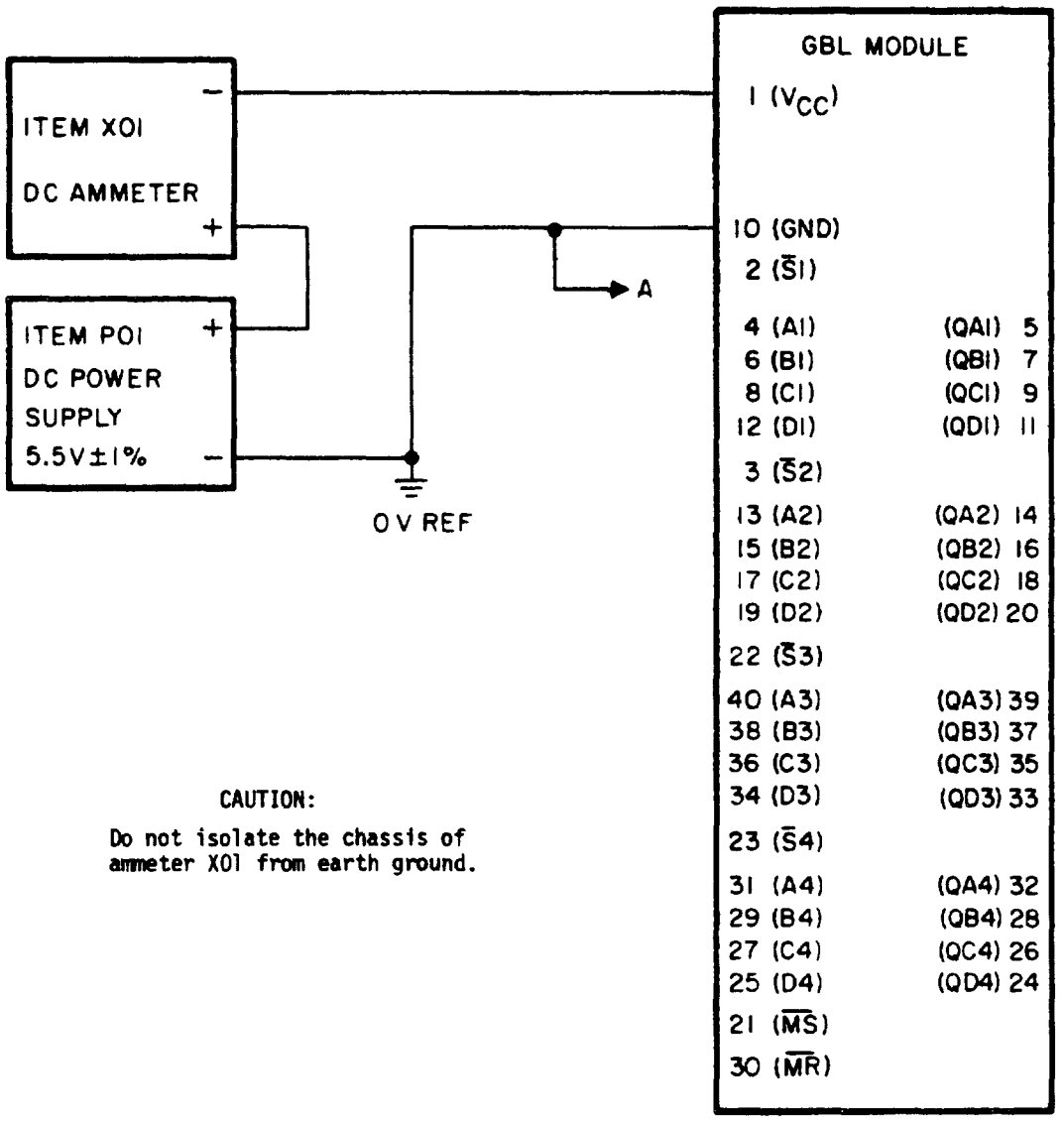


FIGURE 13. Power supply current (all outputs low level) test circuit for table IX.

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TABLE X. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect					Initial limits 25 ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
L1	QA1 Transition time (t_{TLH}), C_L = 50 pF	4	21	5	2	Open	5	40	ns	14, 15
L2	QB1	6		7	3					
L3	QC1	8		9						
L4	QD1	12		11						
L5	QA2	13		14						
L6	QB2	15		16						
L7	QC2	17		18						
L8	QD2	19		20						
L9	QA3	40		39	22					
L10	QB3	38		37						
L11	QC3	36		35						
L12	QD3	34		33						
L13	QA4	31		32	23					
L14	QB4	29		28						
L15	QC4	27		26						
L16	QD4	25		24						
M1	QA1 Transition time (t_{TLH}), C_L = 150 pF	4		5	2			60		
M2	QB1	6		7						
M3	QC1	8		9						
M4	QD1	12		11						

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TABLE X. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M5	QA2	Transition time (t_{TLH}), C_L = 150 pF	13	21	14	3	Open	5	60	ns	14, 15
M6	QB2		15		16						
M7	QC2		17		18						
M8	QD2		19		20						
M9	QA3		40		39	22					
M10	QB3		38		37						
M11	QC3		36		35						
M12	QD3		34		33						
M13	QA4		31		32	23					
M14	QB4		29		28						
M15	QC4		27		26						
M16	QD4		25		24						
N1	QA1	Transition time (t_{THL}), C_L = 50 pF	4		5	2		2	30		
N2	QB1		6		7						
N3	QC1		8		9						
N4	QD1		12		11						

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TABLE X. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin		Min	Max		
N5	QA2 Transition time (t_{THL}), C_L = 50 pF	13	21	14	3	Open		2	30	ns	14, 15
N6	QB2	15		16							
N7	QC2	17		18							
N8	QD2	19		20							
N9	QA3	40		39	22						
N10	QB3	38		37							
N11	QC3	36		35							
N12	QD3	34		33							
N13	QA4	31		32	23						
N14	QB4	29		28							
N15	QC4	27		26							
N16	QD4	25		24							
P1	QA1 Transition time (t_{THL}), C_L = 150 pF	4		5	2				40		
P2	QB1	6		7							
P3	QC1	8		9							
P4	QD1	12		11							

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TABLE X. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P5	QA2	Transition time (t_{THL}), C_L = 150 pF	13	21	14	3	Open	2	40	ns	14, 15
P6	QB2		15		16						
P7	QC2		17		18						
P8	QD2		19		20						
P9	QA3		40		39	22					
P10	QB3		38		37						
P11	QC3		36		35						
P12	QD3		34		33						
P13	QA4		31		32	23					
P14	QB4		29		28						
P15	QC4		27		26						
P16	QD4		25		24						

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TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q1	A1 to QA1	Propagation delay time (t_{PLH}), C_L = 50 pF	4	Open	5	2, 21	Open	10	60	ns	14, 15
Q2	B1 to QB1		6		7						
Q3	C1 to QC1		8		9						
Q4	D1 to QD1		12		11						
Q5	A2 to QA2		13		14	3, 21					
Q6	B2 to QB2		15		16						
Q7	C2 to QC2		17		18						
Q8	D2 to QD2		19		20						
Q9	A3 to QA3		40		39	22, 21					
Q10	B3 to QB3		38		37						
Q11	C3 to QC3		36		35						
Q12	D3 to QD3		34		33						
Q13	A4 to QA4		31		32	23, 21					
Q14	B4 to QB4		29		28						
Q15	C4 to QC4		27		26						
Q16	D4 to QD4		25		24						
R1	A1 to QA1	Propagation delay time (t_{PLH}), C_L = 150 pF	4		5	2, 21			75		
R2	B1 to QB1		6		7						
R3	C1 to QC1		8		9						
R4	D1 to QD1		12		11						

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TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R5	A2 to QA2	Propagation delay time (t_{PLH}), C_L = 150 pF	13	Open	14	3, 21	Open	10	75	ns	14, 15
R6	B2 to QB2	↓	15	↓	16	↓	↓	↓	↓	↓	↓
R7	C2 to QC2	↓	17	↓	18	↓	↓	↓	↓	↓	↓
R8	D2 to QD2	↓	19	↓	20	↓	↓	↓	↓	↓	↓
R9	A3 to QA3	↓	40	↓	39	22, 21	↓	↓	↓	↓	↓
R10	B3 to QB3	↓	38	↓	37	↓	↓	↓	↓	↓	↓
R11	C3 to QC3	↓	36	↓	35	↓	↓	↓	↓	↓	↓
R12	D3 to QD3	↓	34	↓	33	↓	↓	↓	↓	↓	↓
R13	A4 to QA4	↓	31	↓	32	23, 21	↓	↓	↓	↓	↓
R14	B4 to QB4	↓	29	↓	28	↓	↓	↓	↓	↓	↓
R15	C4 to QC4	↓	27	↓	26	↓	↓	↓	↓	↓	↓
R16	D4 to QD4	↓	25	↓	24	↓	↓	↓	↓	↓	↓
S1	A1 to QA1	Propagation delay time (t_{PHL}), C_L = 50 pF	4	↓	5	2, 21	↓	↓	65	↓	↓
S2	B1 to QB1	↓	6	↓	7	↓	↓	↓	↓	↓	↓
S3	C1 to QC1	↓	8	↓	9	↓	↓	↓	↓	↓	↓
S4	D1 to QD1	↓	12	↓	11	↓	↓	↓	↓	↓	↓

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TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
S5	A2 to QA2 Propagation delay time (t_{PHL}), C_L = 50 pF	13	Open	14	3, 21	Open	10	65	ns	14, 15
S6	B2 to QB2	15		16						
S7	C2 to QC2	17		18						
S8	D2 to QD2	19		20						
S9	A3 to QA3	40		39	22, 21					
S10	B3 to QB3	38		37						
S11	C3 to QC3	36		35						
S12	D3 to QD3	34		33						
S13	A4 to QA4	31		32	23, 21					
S14	B4 to QB4	29		28						
S15	C4 to QC4	27		26						
S16	D4 to QD4	25		24						
T1	A1 to QA1 Propagation delay time (t_{PHL}), C_L = 150 pF	4		5	2, 21			80		
T2	B1 to QB1	6		7						
T3	C1 to QC1	8		9						
T4	D1 to QD1	12		11						
T5	A2 to QA2	13		14	3, 21					
T6	B2 to QB2	15		16						
T7	C2 to QC2	17		18						
T8	D2 to QD2	19		20						

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TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T9	A3 to QA3	Propagation delay time (t_{PHL}), C_L = 150 pF	40	Open	39	22, 21	Open	10	80	ns	14, 15
T10	B3 to QB3		38		37						
T11	C3 to QC3		36		35						
T12	D3 to QD3		34		33						
T13	A4 to QA4		31		32	23, 21					
T14	B4 to QB4		29		28						
T15	C4 to QC4		27		26						
T16	D4 to QD4		25		24						
U1	$\overline{MS}$ to QA1	Propagation delay time (t_{PLH}), C_L = 50 pF	4	21	5	2					
U2	$\overline{MS}$ to QB1		6		7						
U3	$\overline{MS}$ to QC1		8		9						
U4	$\overline{MS}$ to QD1		12		11						
U5	$\overline{MS}$ to QA2		13		14	3					
U6	$\overline{MS}$ to QB2		15		16						
U7	$\overline{MS}$ to QC2		17		18						
U8	$\overline{MS}$ to QD2		19		20						
U9	$\overline{MS}$ to QA3		40		39	22					
U10	$\overline{MS}$ to QB3		38		37						
U11	$\overline{MS}$ to QC3		36		35						
U12	$\overline{MS}$ to QD3		34		33						
U13	$\overline{MS}$ to QA4		31		32	23					
U14	$\overline{MS}$ to QB4		29		28						
U15	$\overline{MS}$ to QC4		27		26						
U16	$\overline{MS}$ to QD4		25		24						

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TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V1	$\overline{MS}$ to QA1	Propagation delay time (t_{PLH}), C_L = 150 pF	4	21	5	2	Open	10	95	ns	14, 15
V2	$\overline{MS}$ to QB1		6		7						
V3	$\overline{MS}$ to QC1		8		9						
V4	$\overline{MS}$ to QD1		12		11						
V5	$\overline{MS}$ to QA2		13		14	3					
V6	$\overline{MS}$ to QB2		15		16						
V7	$\overline{MS}$ to QC2		17		18						
V8	$\overline{MS}$ to QD2		19		20						
V9	$\overline{MS}$ to QA3		40		39	22					
V10	$\overline{MS}$ to QB3		38		37						
V11	$\overline{MS}$ to QC3		36		35						
V12	$\overline{MS}$ to QD3		34		33						
V13	$\overline{MS}$ to QA4		31		32	23					
V14	$\overline{MS}$ to QB4		29		28						
V15	$\overline{MS}$ to QC4		27		26						
V16	$\overline{MS}$ to QD4		25		24						
W1	$\overline{MS}$ to QA1	Propagation delay time (t_{PHL}), C_L = 50 pF	4		5	2			65		
W2	$\overline{MS}$ to QB1		6		7						
W3	$\overline{MS}$ to QC1		8		9						
W4	$\overline{MS}$ to QD1		12		11						
W5	$\overline{MS}$ to QA2		13		14	3					

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TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W6	$\overline{MS}$ to QB2	Propagation delay time (t_{PHL}), C_L = 50 pF	15	21	16	3	Open	10	65	ns	14, 15
W7	$\overline{MS}$ to QC2		17		18	3					
W8	$\overline{MS}$ to QD2		19		20	3					
W9	$\overline{MS}$ to QA3		40		39	22					
W10	$\overline{MS}$ to QB3		38		37						
W11	$\overline{MS}$ to QC3		36		35						
W12	$\overline{MS}$ to QD3		34		33						
W13	$\overline{MS}$ to QA4		31		32	23					
W14	$\overline{MS}$ to QB4		29		28						
W15	$\overline{MS}$ to QC4		27		26						
W16	$\overline{MS}$ to QD4		25		24						
X1	MS to QA1	Propagation delay time (t_{PHL}), C_L = 150 pF	4		5	2			80		
X2	$\overline{MS}$ to QB1		6		7						
X3	$\overline{MS}$ to QC1		8		9						
X4	$\overline{MS}$ to QD1		12		11						
X5	$\overline{MS}$ to QA2		13		14	3					
X6	MS to QB2		15		16	3					
X7	$\overline{MS}$ to QC2		17		18	3					

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TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
X8	$\overline{MS}$ to QD2	Propagation delay time (t_{PHL}), C_L = 150 pF	19	21	20	3	Open	10	80	ns	14, 15
X9	$\overline{MS}$ to QA3		40		39	22					
X10	$\overline{MS}$ to QB3		38		37						
X11	$\overline{MS}$ to QC3		36		35						
X12	$\overline{MS}$ to QD3		34		33						
X13	$\overline{MS}$ to QA4		31		32	23					
X14	$\overline{MS}$ to QB4		29		28						
X15	$\overline{MS}$ to QC4		27		26						
X16	$\overline{MS}$ to QD4		25		24						
Y1	S1 to QA1	Propagation delay time (t_{PLH}), C_L = 50 pF	4	2	5	21					
Y2	$\overline{S}$1 to QB1		6		7						
Y3	$\overline{S}$1 to QC1		8		9						
Y4	$\overline{S}$1 to QD1		12		11						
Y5	$\overline{S}$2 to QA2		13	3	14						
Y6	$\overline{S}$2 to QB2		15	3	16						
Y7	$\overline{S}$2 to QC2		17	3	18						

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TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y8	$\overline{S}2$ to QD2	Propagation delay time (t_{PLH}), C_L = 50 pF	19	3	20	21	Open	10	80	ns	14, 15
Y9	$\overline{S}3$ to QA3	↓	40	22	39	↓	↓	↓	↓	↓	↓
Y10	$\overline{S}3$ to QB3	↓	38	↓	37	↓	↓	↓	↓	↓	↓
Y11	$\overline{S}3$ to QC3	↓	36	↓	35	↓	↓	↓	↓	↓	↓
Y12	$\overline{S}3$ to QD3	↓	34	↓	33	↓	↓	↓	↓	↓	↓
Y13	$\overline{S}4$ to QA4	↓	31	23	32	↓	↓	↓	↓	↓	↓
Y14	$\overline{S}4$ to QB4	↓	29	↓	28	↓	↓	↓	↓	↓	↓
Y15	$\overline{S}4$ to QC4	↓	27	↓	26	↓	↓	↓	↓	↓	↓
Y16	$\overline{S}4$ to QD4	↓	25	↓	24	↓	↓	↓	↓	↓	↓
Z1	$\overline{S}1$ to QA1	Propagation delay time (t_{PLH}), C_L = 150 pF	4	2	5	↓	↓	↓	95	↓	↓
Z2	$\overline{S}1$ to QB1	↓	6	↓	7	↓	↓	↓	↓	↓	↓
Z3	$\overline{S}1$ to QC1	↓	8	↓	9	↓	↓	↓	↓	↓	↓
Z4	$\overline{S}1$ to QD1	↓	12	↓	11	↓	↓	↓	↓	↓	↓
Z5	$\overline{S}2$ to QA2	↓	13	3	14	↓	↓	↓	↓	↓	↓
Z6	$\overline{S}2$ to QB2	↓	15	3	16	↓	↓	↓	↓	↓	↓
Z7	$\overline{S}2$ to QC2	↓	17	3	18	↓	↓	↓	↓	↓	↓

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TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z8	$\overline{S}2$ to QD2	Propagation delay time (t_{PLH}), C_L = 150 pF	19	3	20	21	Open	10	95	ns	14, 15
Z9	$\overline{S}3$ to QA3		40	22	39						
Z10	$\overline{S}3$ to QB3		38		37						
Z11	$\overline{S}3$ to QC3		36		35						
Z12	$\overline{S}3$ to QD3		34		33						
Z13	$\overline{S}4$ to QA4		31	23	32						
Z14	$\overline{S}4$ to QB4		29		28						
Z15	$\overline{S}4$ to QC4		27		26						
Z16	$\overline{S}4$ to QD4		25		24						
AA1	$\overline{S}1$ to QA1	Propagation delay time (t_{PHL}), C_L = 50 pF	4	2	5	21			65		
AA2	$\overline{S}1$ to QB1		6		7						
AA3	$\overline{S}1$ to QC1		8		9						
AA4	$\overline{S}1$ to QD1		12		11						
AA5	$\overline{S}2$ to QA2		13	3	14						
AA6	$\overline{S}2$ to QB2		15	3	16						
AA7	$\overline{S}2$ to QC2		17	3	18						

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TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA8	$\overline{S}2$ to QD2	Propagation delay time (t_{PHL}), C_L = 50 pF	19	3	20	21	Open	10	65	ns	14, 15
AA9	$\overline{S}3$ to QA3	↓	40	22	39	↓	↓	↓	↓	↓	↓
AA10	$\overline{S}3$ to QB3	↓	38	↓	37	↓	↓	↓	↓	↓	↓
AA11	$\overline{S}3$ to QC3	↓	36	↓	35	↓	↓	↓	↓	↓	↓
AA12	$\overline{S}3$ to QD3	↓	34	↓	33	↓	↓	↓	↓	↓	↓
AA13	$\overline{S}4$ to QA4	↓	31	23	32	↓	↓	↓	↓	↓	↓
AA14	$\overline{S}4$ to QB4	↓	29	↓	28	↓	↓	↓	↓	↓	↓
AA15	$\overline{S}4$ to QC4	↓	27	↓	26	↓	↓	↓	↓	↓	↓
AA16	$\overline{S}4$ to QD4	↓	25	↓	24	↓	↓	↓	↓	↓	↓
BB1	$\overline{S}1$ to QA1	Propagation delay time (t_{PHL}), C_L = 150 pF	4	2	5	↓	↓	↓	80	↓	↓
BB2	$\overline{S}1$ to QB1	↓	6	↓	7	↓	↓	↓	↓	↓	↓
BB3	$\overline{S}1$ to QC1	↓	8	↓	9	↓	↓	↓	↓	↓	↓
BB4	$\overline{S}1$ to QD1	↓	12	↓	11	↓	↓	↓	↓	↓	↓
BB5	$\overline{S}2$ to QA2	↓	13	3	14	↓	↓	↓	↓	↓	↓
BB6	$\overline{S}2$ to QB2	↓	15	3	16	↓	↓	↓	↓	↓	↓
BB7	$\overline{S}2$ to QC2	↓	17	3	18	↓	↓	↓	↓	↓	↓

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TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB8	$\overline{S2}$ to QD2 Propagation delay time (t_{PHL}), C_L = 150 pF	19	3	20	21	Open	10	80	ns	14, 15
BB9	$\overline{S3}$ to QA3	40	22	39						
BB10	$\overline{S3}$ to QB3	38		37						
BB11	$\overline{S3}$ to QC3	36		35						
BB12	$\overline{S3}$ to QD3	34		33						
BB13	$\overline{S4}$ to QA4	31	23	32						
BB14	$\overline{S4}$ to QB4	29		28						
BB15	$\overline{S4}$ to QC4	27		26						
BB16	$\overline{S4}$ to QD4	25		24						

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* TABLE XII. Setup and hold time check test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC1	A1 to $\overline{S1}$ Setup time check	4	2	5	21	Open	175	205	ns	14, 16
CC2	B1 to $\overline{S1}$	6		7						
CC3	C1 to $\overline{S1}$	8		9						
CC4	D1 to $\overline{S1}$	12		11						
CC5	A2 to $\overline{S2}$	13	3	14						
CC6	B2 to $\overline{S2}$	15		16						
CC7	C2 to $\overline{S2}$	17		18						
CC8	D2 to $\overline{S2}$	19		20						
CC9	A3 to $\overline{S3}$	40	22	39						
CC10	B3 to $\overline{S3}$	38		37						
CC11	C3 to $\overline{S3}$	36		35						
CC12	D3 to $\overline{S3}$	34		33						
CC13	A4 to $\overline{S4}$	31	23	32						
CC14	B4 to $\overline{S4}$	29		28						
CC15	C4 to $\overline{S4}$	27		26						
CC16	D4 to $\overline{S4}$	25		24						

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* TABLE XII. Setup and hold time check test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
DD1	A1 to $\overline{S1}$ Hold time check	4	2	5	21	Open	185	215	ns	14, 17
DD2	B1 to $\overline{S1}$	6		7						
DD3	C1 to $\overline{S1}$	8		9						
DD4	D1 to $\overline{S1}$	12		11						
DD5	A2 to $\overline{S2}$	13	3	14						
DD6	B2 to $\overline{S2}$	15		16						
DD7	C2 to $\overline{S2}$	17		18						
DD8	D2 to $\overline{S2}$	19		20						
DD9	A3 to $\overline{S3}$	40	22	39						
DD10	B3 to $\overline{S3}$	38		37						
DD11	C3 to $\overline{S3}$	36		35						
DD12	D3 to $\overline{S3}$	34		33						
DD13	A4 to $\overline{S4}$	31	23	32						
DD14	B4 to $\overline{S4}$	29		28						
DD15	C4 to $\overline{S4}$	27		26						
DD16	D4 to $\overline{S4}$	25		24						

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* TABLE XII. Setup and hold time check test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
EE1	A1 to $\overline{MS}$ Setup time check	4	21	5	2	Open	175	205	ns	14, 16
EE2	B1 to $\overline{MS}$	6	↓	7	↓	↓	↓	↓	↓	↓
EE3	C1 to $\overline{MS}$	8	↓	9	↓	↓	↓	↓	↓	↓
EE4	D1 to $\overline{MS}$	12	↓	11	↓	↓	↓	↓	↓	↓
EE5	A2 to $\overline{MS}$	13	↓	14	3	↓	↓	↓	↓	↓
EE6	B2 to $\overline{MS}$	15	↓	16	↓	↓	↓	↓	↓	↓
EE7	C2 to $\overline{MS}$	17	↓	18	↓	↓	↓	↓	↓	↓
EE8	D2 to $\overline{MS}$	19	↓	20	↓	↓	↓	↓	↓	↓
EE9	A3 to $\overline{MS}$	40	↓	39	22	↓	↓	↓	↓	↓
EE10	B3 to $\overline{MS}$	38	↓	37	↓	↓	↓	↓	↓	↓
EE11	C3 to $\overline{MS}$	36	↓	35	↓	↓	↓	↓	↓	↓
EE12	D3 to $\overline{MS}$	34	↓	33	↓	↓	↓	↓	↓	↓
EE13	A4 to $\overline{MS}$	31	↓	32	23	↓	↓	↓	↓	↓
EE14	B4 to $\overline{MS}$	29	↓	28	↓	↓	↓	↓	↓	↓
EE15	C4 to $\overline{MS}$	27	↓	26	↓	↓	↓	↓	↓	↓
EE16	D4 to $\overline{MS}$	25	↓	24	↓	↓	↓	↓	↓	↓

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TABLE XII. Setup and hold time check test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
FF1	A1 to MS Hold time check	4	21	5	2	Open	185	215	ns	14, 17
FF2	B1 to MS	6		7						
FF3	C1 to MS	8		9						
FF4	D1 to MS	12		11						
FF5	A2 to MS	13		14	3					
FF6	B2 to MS	15		16						
FF7	C2 to MS	17		18						
FF8	D2 to MS	19		20						
FF9	A3 to MS	40		39	22					
FF10	B3 to MS	38		37						
FF11	C3 to MS	36		35						
FF12	D3 to MS	34		33						
FF13	A4 to MS	31		32	23					
FF14	B4 to MS	29		28						
FF15	C4 to MS	27		26						
FF16	D4 to MS	25		24						

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★ TABLE XIII. Master reset delay time test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
GG1	Master reset delay time, C_L = 50 pF	2	30	5	21	4,6,8,12	10	65	ns	14,18
GG2				7						
GG3				9						
GG4				11						
GG5		3		14		13,15,17,19				
GG6				16						
GG7				18						
GG8				20						
GG9		22		39		40,38,36,34				
GG10				37						
GG11				35						
GG12				33						
GG13		23		32		31,29,27,25				
GG14				28						
GG15				26						
GG16				24						
GG17	Master reset delay time, C_L = 50 pF, PWC	2		5		4,6,8,12	400	600		
GG18				7						
GG19				9						
GG20				11						

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TABLE XIII. Master reset delay time test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
GG21	Master reset delay time, C_L = 50 pF, PWC	3	30	14	21	13,15,17,19	400	600	ns	14,18
GG22				16						
GG23				18						
GG24				20						
GG25		22		39		40,38,36,34				
GG26				37						
GG27				35						
GG28				33						
GG29		23		32		31,29,27,25				
GG30				28						
GG31				26						
GG32				24						
HH1	Master reset delay time, C_L = 150 pF	2		5		4,6,8,12	10	80		
HH2				7						
HH3				9						
HH4				11						
HH5		3		14		13,15,17,19				
HH6				16						
HH7				18						
HH8				20						

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TABLE XIII. Master reset delay time test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
HH9	Master reset delay time, C_L = 150 pF	22	30	39	21	40,38,36,34	10	80	ns	14, 18
HH10				37						
HH11				35						
HH12				33						
HH13		23		32		31,29,27,25				
HH14				28						
HH15				26						
HH16				24						
HH17	Master reset delay time, C_L = 150 pF, PWC	2		5		4,6,8,12	400	600		
HH18				7						
HH19				9						
HH20				11						
HH21		3		14		13,15,17,19				
HH22				16						
HH23				18						
HH24				20						
HH25		22		39		40,38,36,34				
HH26				37						
HH27				35						
HH28				33						

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TABLE XIII. Master reset delay time test - Continued.

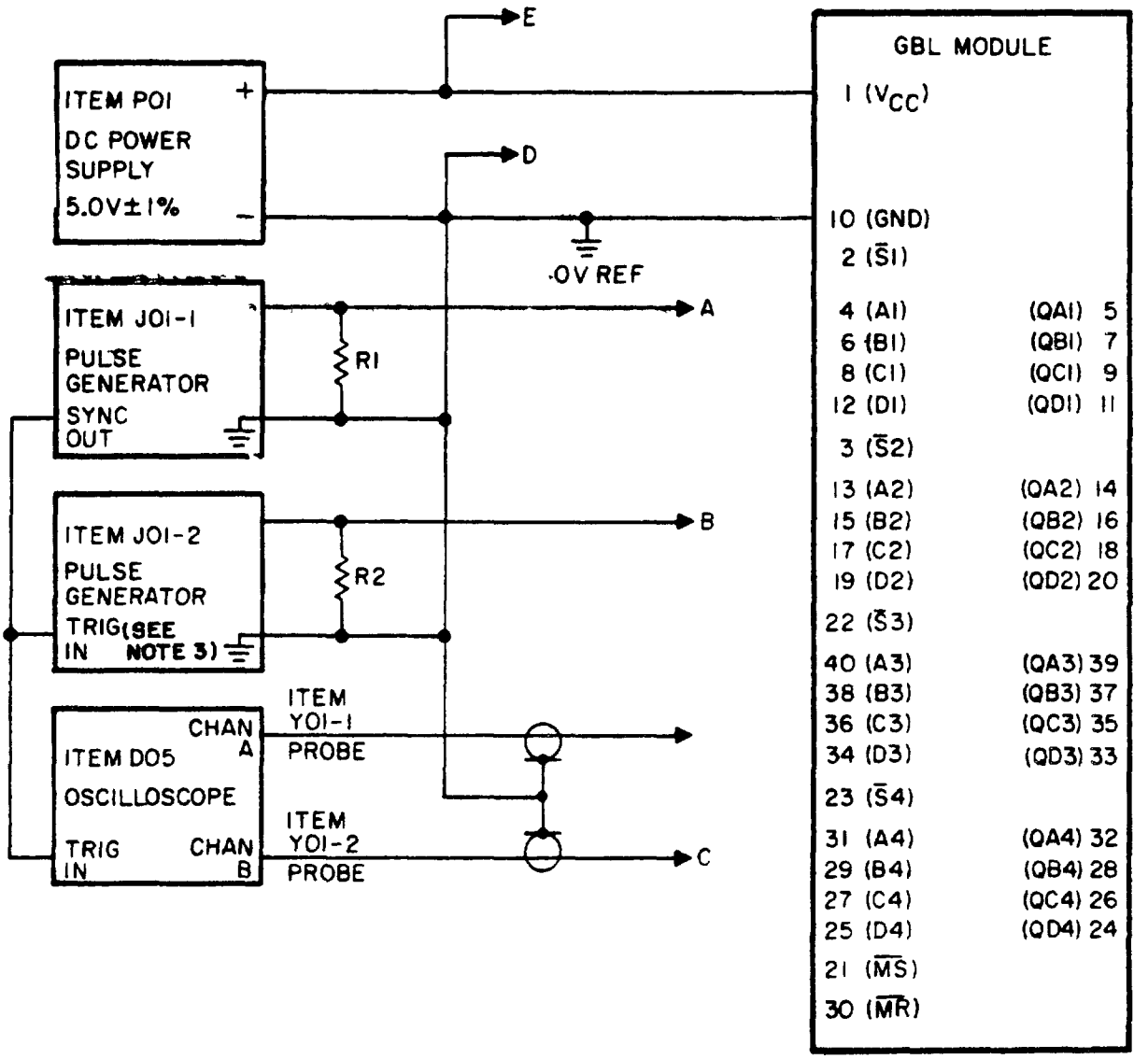
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
HH29	Master reset delay time, C_L = 150 pF, PWC	23	30	32	21	31,29,27,25	400	600	ns	14,18
HH30	↓	↓	↓	28	↓	↓	↓	↓	↓	↓
HH31				26						
HH32				24						

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TABLE XIV. Master reset recovery time check test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
JJ1	Master reset recovery time check	21	30	5	2	Open	180	220	ns	14, 19
JJ2				7						
JJ3				9						
JJ4				11						
JJ5				14	3					
JJ6				16						
JJ7				18						
JJ8				20						
JJ9				39	22					
JJ10				37						
JJ11				35						
JJ12				33						
JJ13				32	23					
JJ14				28						
JJ15				26						
JJ16				24						

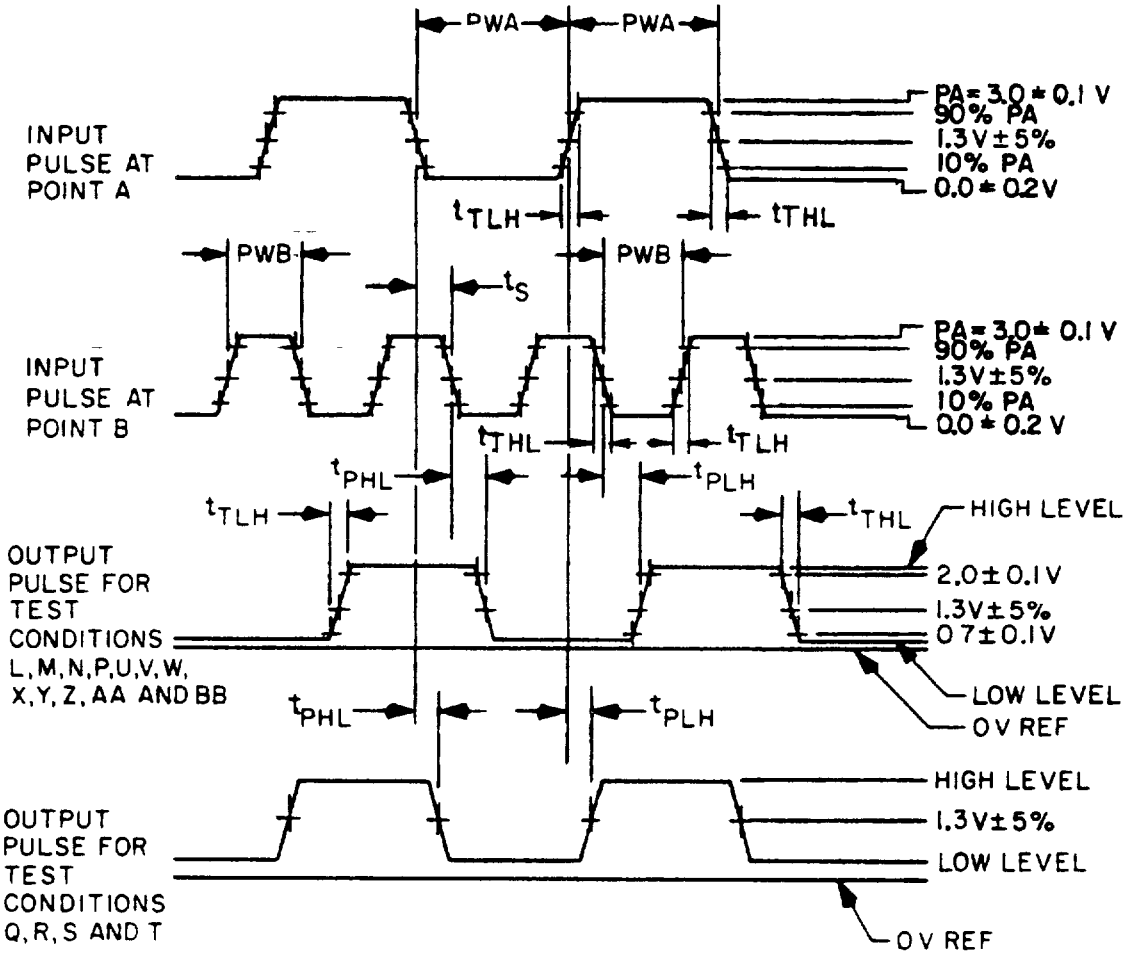
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NOTES.
1. R1 = R2 = 50Ω ±5%, 1 W.
2. Load all outputs of the circuit under test with a load as specified on figure 12.
3 Place item J01-2 in the double pulse mode except for test conditions GG and HH.

FIGURE 14. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PLH} and t_{PHL}), setup and hold time check, master reset delay time, and master reset recovery time check test circuit for tables X, XI, XII, XIII, and XIV.

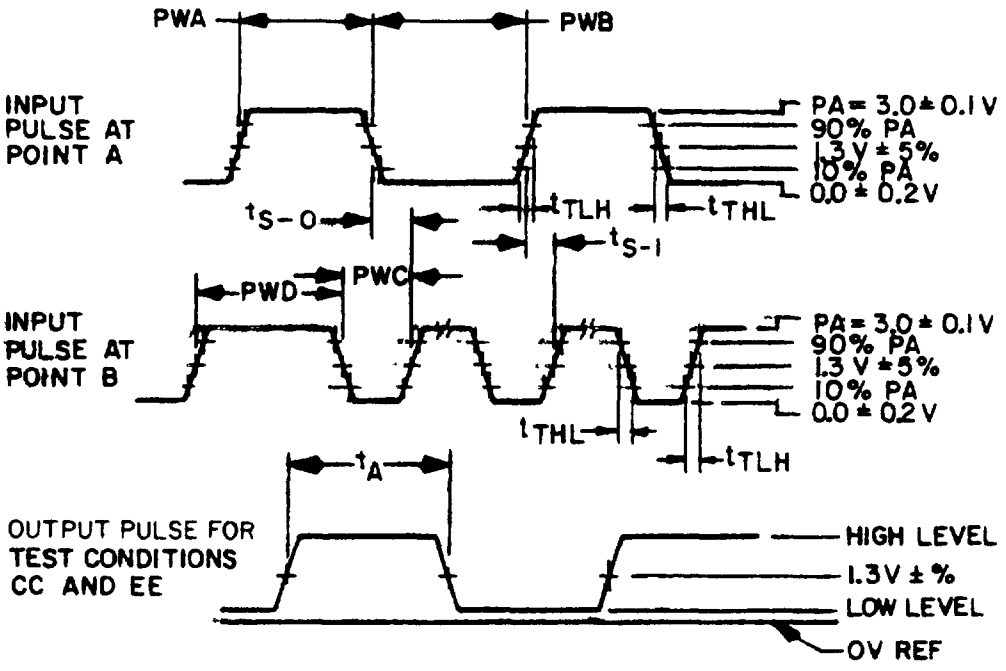
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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PWB = 250 ±25 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   $t_S = 125 \pm 12$ ns.
2. Before making the output pulse measurements, the output levels should be verified to be 0.3 V maximum for the low level and 2.4 V minimum for the high level.
3. All voltage levels are referenced to ground.

* FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables X and XI.

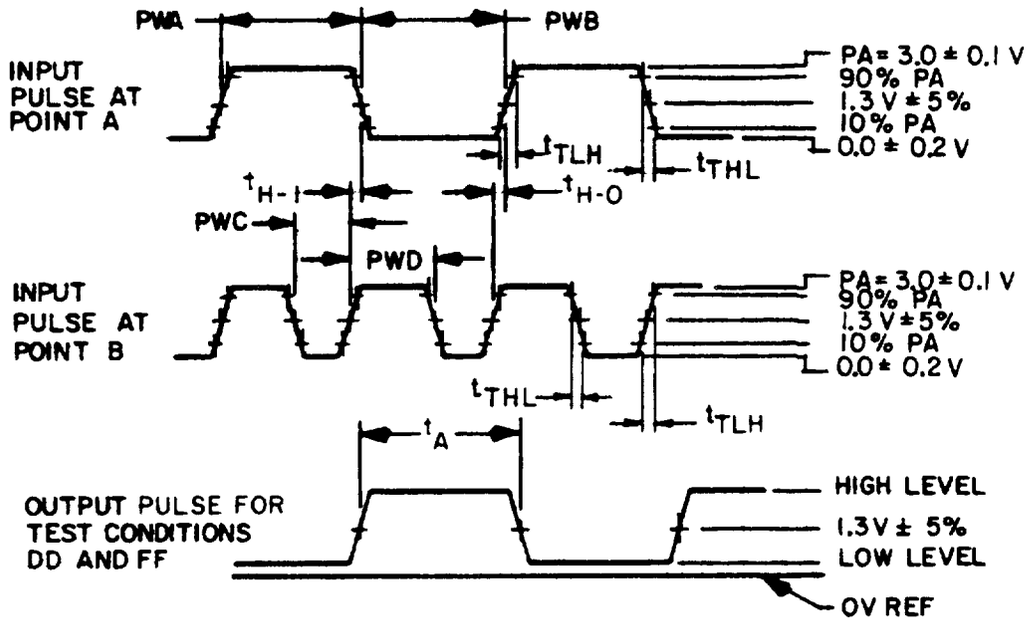
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NOTES:
1. Input pulse parameters:
   PWA = 190 ±10 ns.
   PWB = 210 ±10 ns.
   PWC = 45 $^{+2}_{-0}$ ns.
   PWD = 155 ±10% ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   t_{S-1} = 25 $^{+2}_{-0}$ ns.
   t_{S-0} = 35 $^{+2}_{-0}$ ns.
2. All voltage levels are referenced to ground.

* FIGURE 16. Setup and hold time check test circuit waveforms for table XII.

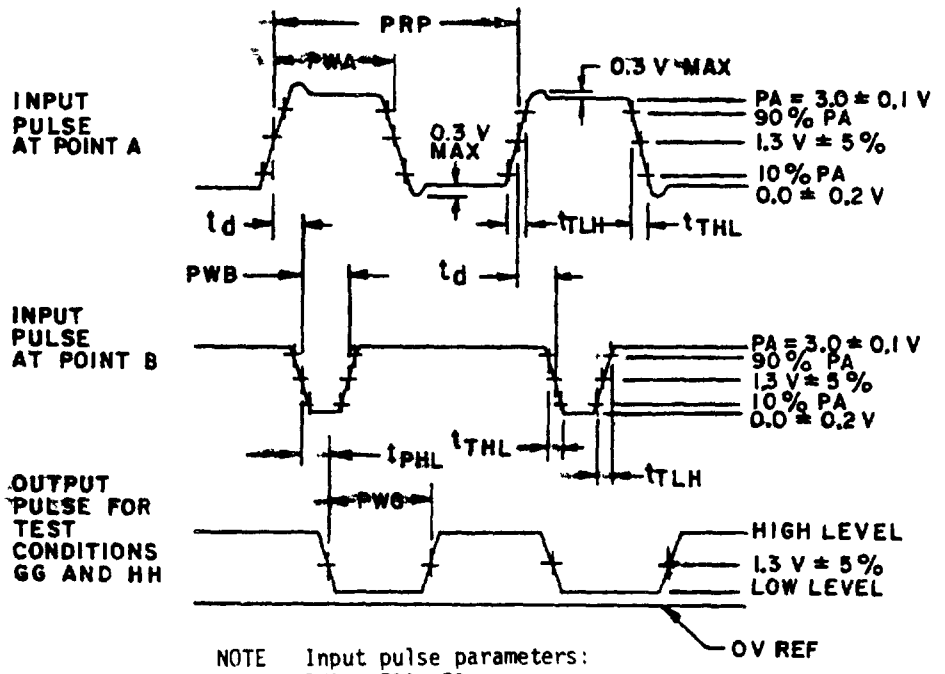
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NOTES:
1. Input pulse parameters:
   PWA = 200 ±10 ns.
   PWB = 200 ±10 ns.
   PWC = 45 $^{+2}_{-0}$ ns.
   PWD = 155 ±10 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   t_{H-1} = 15 $^{+2}_{-0}$ ns.
   t_{H-0} = 15 $^{+2}_{-0}$ ns.
2. All voltage levels are referenced to ground.

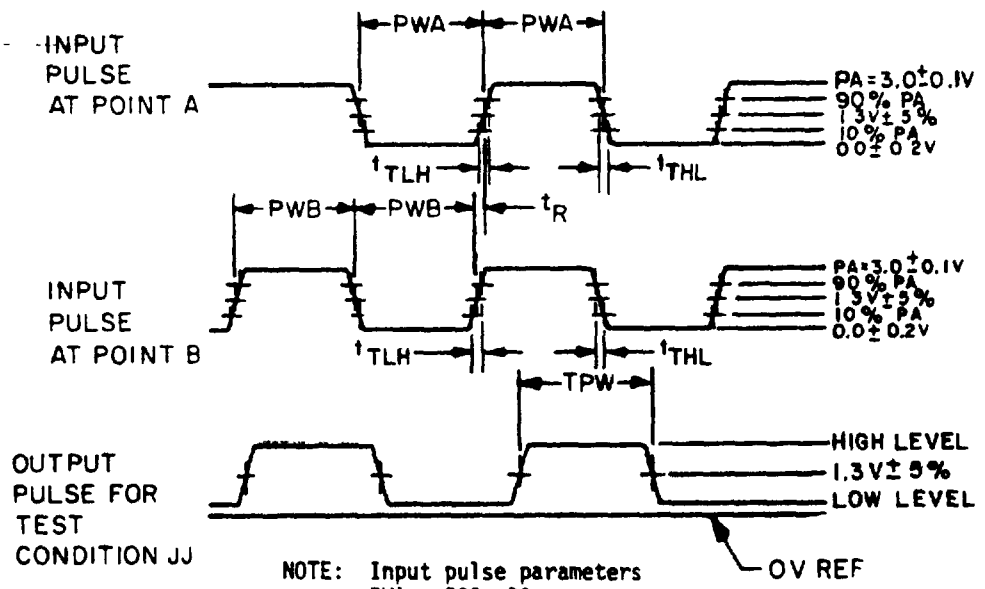
* FIGURE 17. Setup and hold time check test circuit waveforms for table XII.

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NOTE Input pulse parameters:
PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns
t_d = 50 ±5 ns.
PWB = $70\ ^{+2}_{-0}$ ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns

FIGURE 18 Master reset delay time test circuit waveforms for table XIII.



NOTE: Input pulse parameters
PWA = 200 ±20 ns.
PWB = 200 ±20 ns.
$t_R = 30\ ^{+2}_{-0}$ ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.

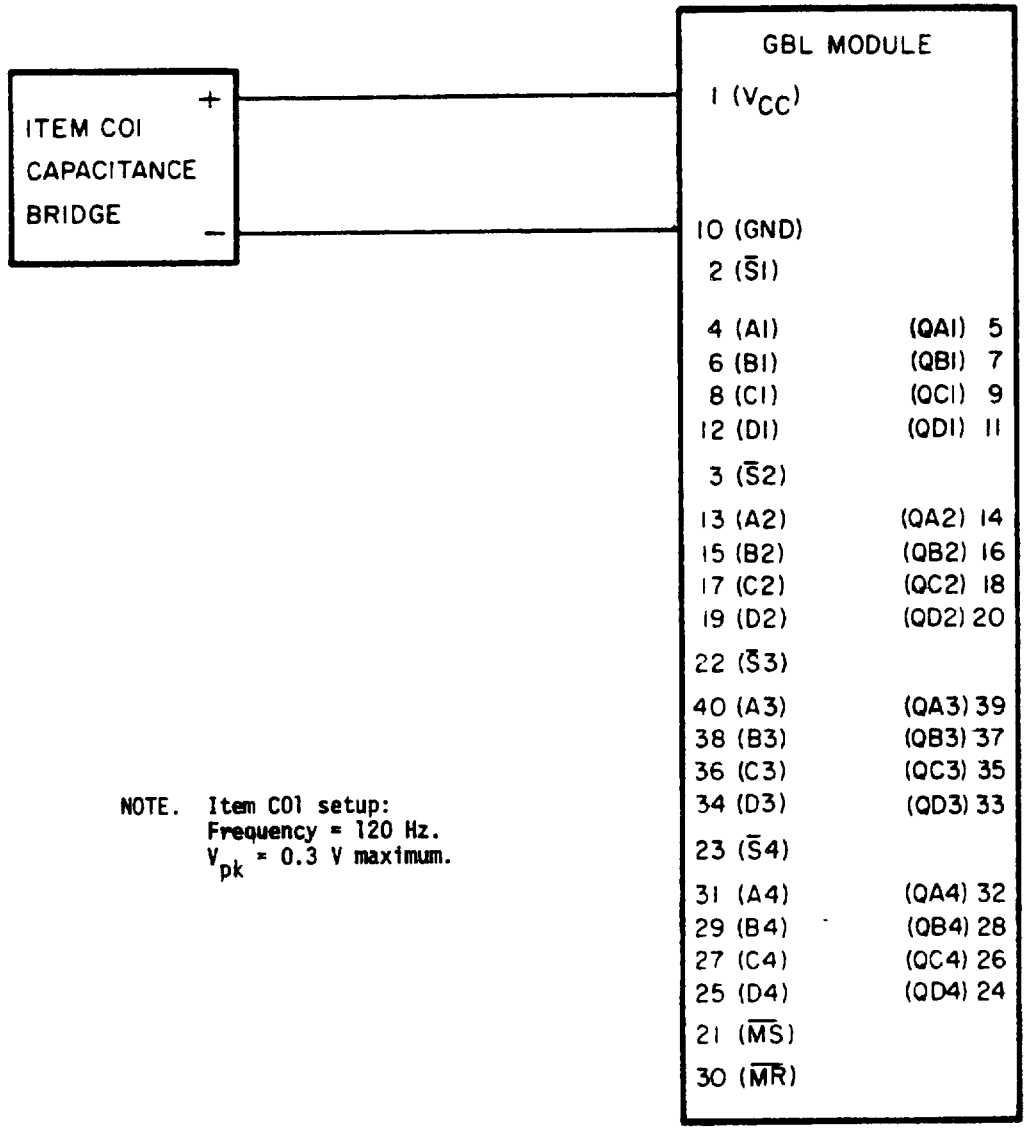
* FIGURE 19. Master reset recovery time check test circuit waveforms for table XIV.

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TABLE XV. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
KK	Power supply filter capacitance	1.6	3.0	µF	20

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NOTE. Item COI setup:
Frequency = 120 Hz.
V_{pk} = 0.3 V maximum.

FIGURE 20. Power supply filter capacitance test circuit for table XV.

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TABLE XVI. Output crosstalk (high-level and low-level) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
LL1	QA1 Output crosstalk, high level	4,30	Open	6,8,12,13,15,17,19,40,38,36, 34,31,29,27,25	5	0	400	mV	21, 22
LL2	QB1	6,30		4,8,12,13,15,17,19,40,38,36, 34,31,29,27,25	7				
LL3	QC1	8,30		4,6,12,13,15,17,19,40,38,36, 34,31,29,27,25	9				
LL4	QD1	12,30		4,6,8,13,15,17,19,40,38,36, 34,31,29,27,25	11				
LL5	QA2	13,30		4,6,8,12,15,17,19,40,38,36, 34,31,29,27,25	14				
LL6	QB2	15,30		4,6,8,12,13,17,19,40,38,36, 34,31,29,27,25	16				
LL7	QC2	17,30		4,6,8,12,13,15,19,40,38,36, 34,31,29,27,25	18				
LL8	QC2	19,30		4,6,8,12,13,15,17,40,38,36, 34,31,29,27,25	20				
LL9	QA3	40,30		4,6,8,12,13,15,17,19,38,36, 34,31,29,27,25	39				
LL10	QB3	38,30		4,6,8,12,13,15,17,19,40,36, 34,31,29,27,25	37				
LL11	QC3	36,30		4,6,8,12,13,15,17,19,40,38, 34,31,29,27,25	35				
LL12	QD3	34,30		4,6,8,12,13,15,17,19,40,38, 36,31,29,27,25	33				
LL13	QA4	31,30		4,6,8,12,13,15,17,19,40,38, 36,34,29,27,25	32				
LL14	QB4	29,30		4,6,8,12,13,15,17,19,40,38, 36,34,31,27,25	28				
LL15	QC4	27,30		4,6,8,12,13,15,17,19,40,38, 36,34,31,29,25	26				
LL16	QD4	25,30		4,6,8,12,13,15,17,19,40,38, 36,34,31,29,27	24				

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TABLE XVI. Output crosstalk (high-level and low-level) test - Continued.

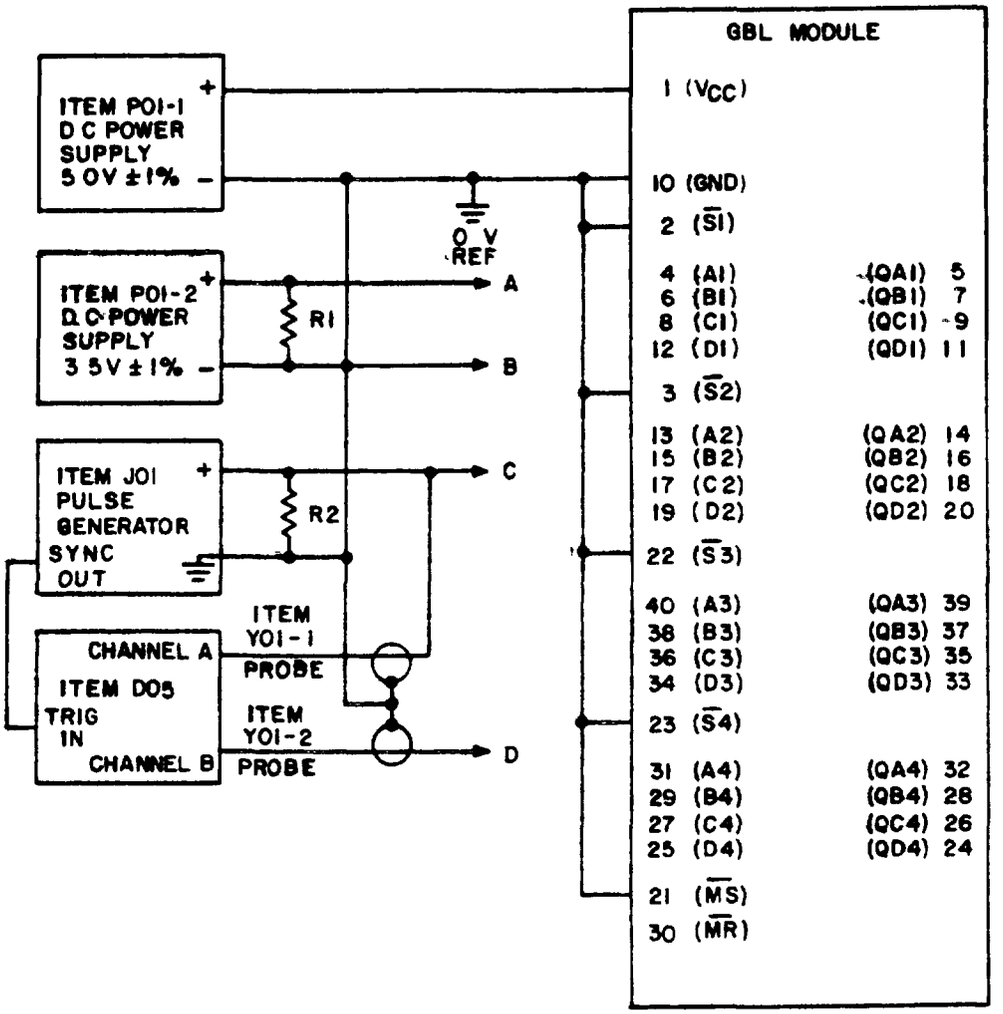
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
MM1	QA1	Output crosstalk, low level	30	4	6,8,12,13,15,17,19,40,38,36, 34,31,29,27,25	5	0	750	mV	21, 22
MM2	QB1			6	4,8,12,13,15,17,19,40,38,36, 34,31,29,27,25	7				
MM3	QC1			8	4,6,12,13,15,17,19,40,38,36, 34,31,29,27,25	9				
MM4	QD1			12	4,6,8,13,15,17,19,40,38,36, 34,31,29,27,25	11				
MM5	QA2			13	4,6,8,12,15,17,19,40,38,36, 34,31,29,27,25	14				
MM6	QB2			15	4,6,8,12,13,17,19,40,38,36, 34,31,29,27,25	16				
MM7	QC2			17	4,6,8,12,13,15,19,40,38,36, 34,31,29,27,25	18				
MM8	QD2			19	4,6,8,12,13,15,17,40,38,36, 34,31,29,27,25	20				
MM9	QA3			40	4,6,8,12,13,15,17,19,38,36, 34,31,29,27,25	39				
MM10	QB3			38	4,6,8,12,13,15,17,19,40,36, 34,31,29,27,25	37				
MM11	QC3			36	4,6,8,12,13,15,17,19,40,38, 34,31,29,27,25	35				
MM12	QD3			34	4,6,8,12,13,15,17,19,40,38, 36,31,29,27,25	33				
MM13	QA4			31	4,6,8,12,13,15,17,19,40,38, 36,34,29,27,25	32				
MM14	QB4			29	4,6,8,12,13,15,17,19,40,38, 36,34,31,27,25	28				
MM15	QC4			27	4,6,8,12,13,15,17,19,40,38, 36,34,31,29,25	26				
MM16	QD4			25	4,6,8,12,13,15,17,19,40,38, 36,34,31,29,27	24				

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TABLE XVI. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
MM17	QA1	Output crosstalk, low level	30	4,6,8,12	13,15,17,19,25,27,29,31,34,36,38,40	5	0	400	mV	21,22
MM18	QB1					7				
MM19	QC1					9				
MM20	QD1					11				
MM21	QA2			13,15,17,19	4,6,8,12,25,27,29,31,34,36,38,40	14				
MM22	QB2					16				
MM23	QC2					18				
MM24	QD2					20				
MM25	QA3			34,36,38,40	4,6,8,12,13,15,17,19,25,27,29,31	39				
MM26	QB3					37				
MM27	QC3					35				
MM28	QD3					33				
MM29	QA4			25,27,29,31	4,6,8,12,13,15,17,19,34,36,38,40	32				
MM30	QB4					28				
MM31	QC4					26				
MM32	QD4					24				

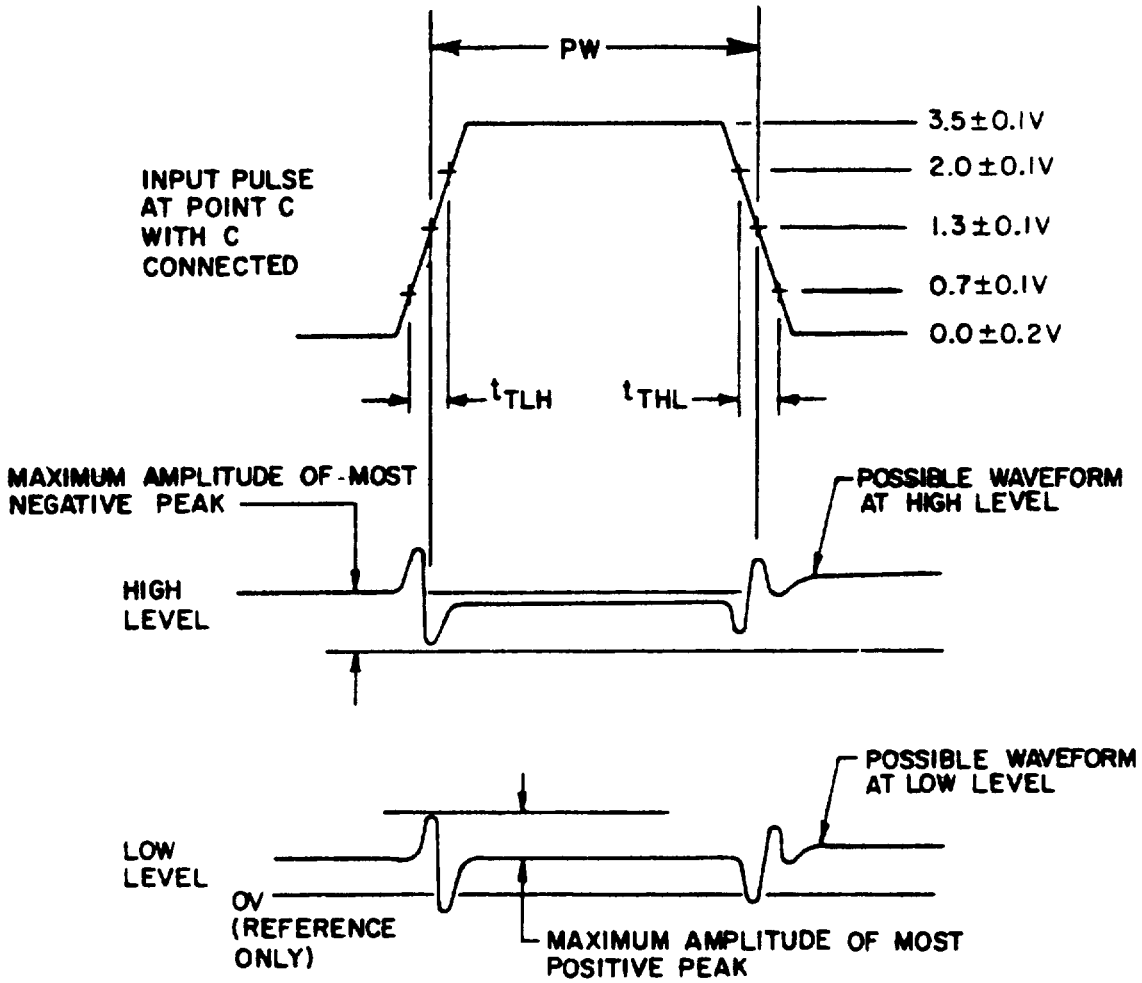
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. Load each output with a load as specified on figure 12.

FIGURE 21. Output crosstalk (high-level and low-level) test circuit for table XVI.

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NOTE: Input pulse parameters:
PW = 200 ±20 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
PRR = 1 MHz ±10%.

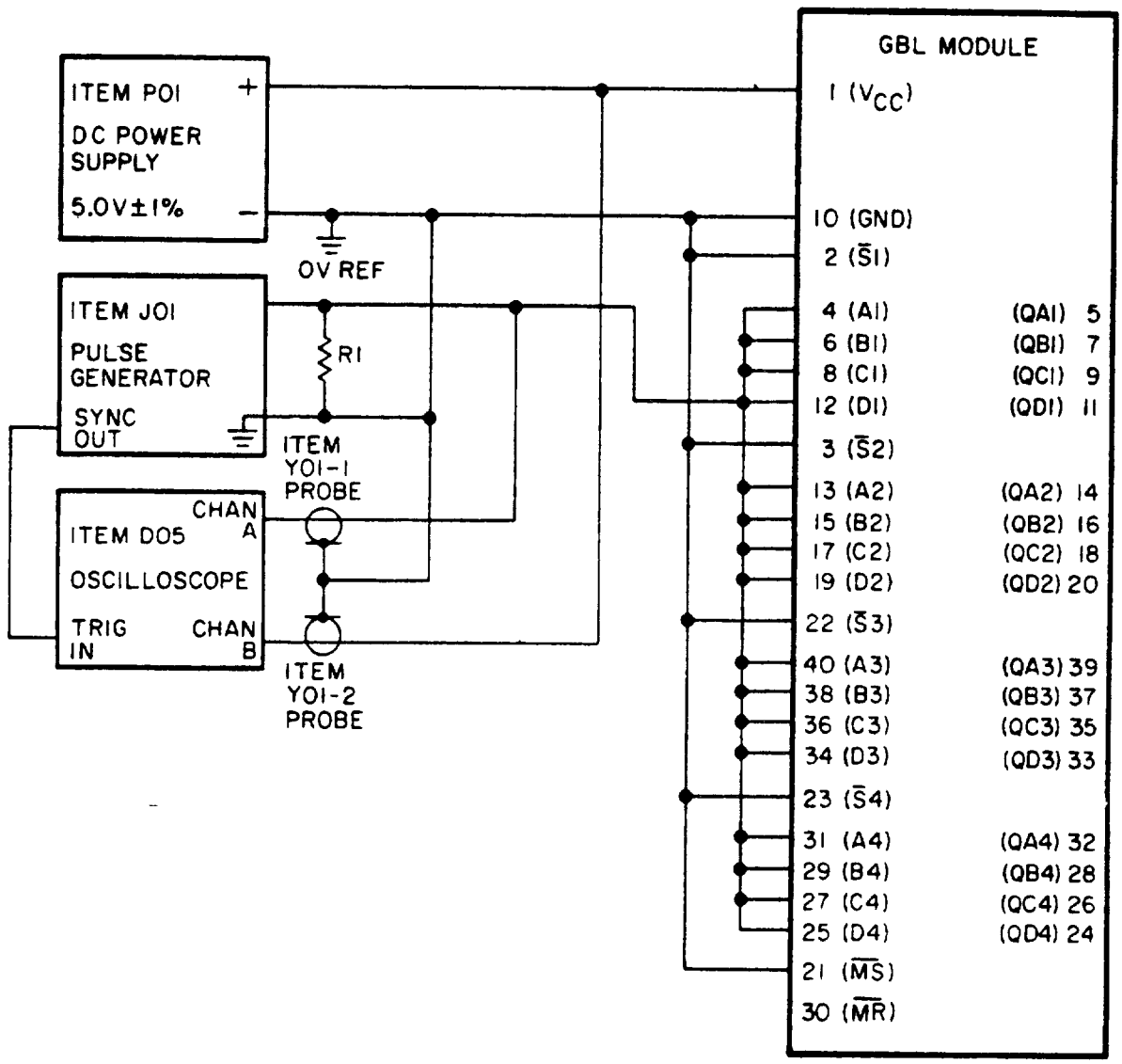
★ FIGURE 22. Output crosstalk (high-level and low-level) test circuit waveforms for table XVI.

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TABLE TABLE XVII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
NN	Power supply transient amplitude	0	300	mV	23,24

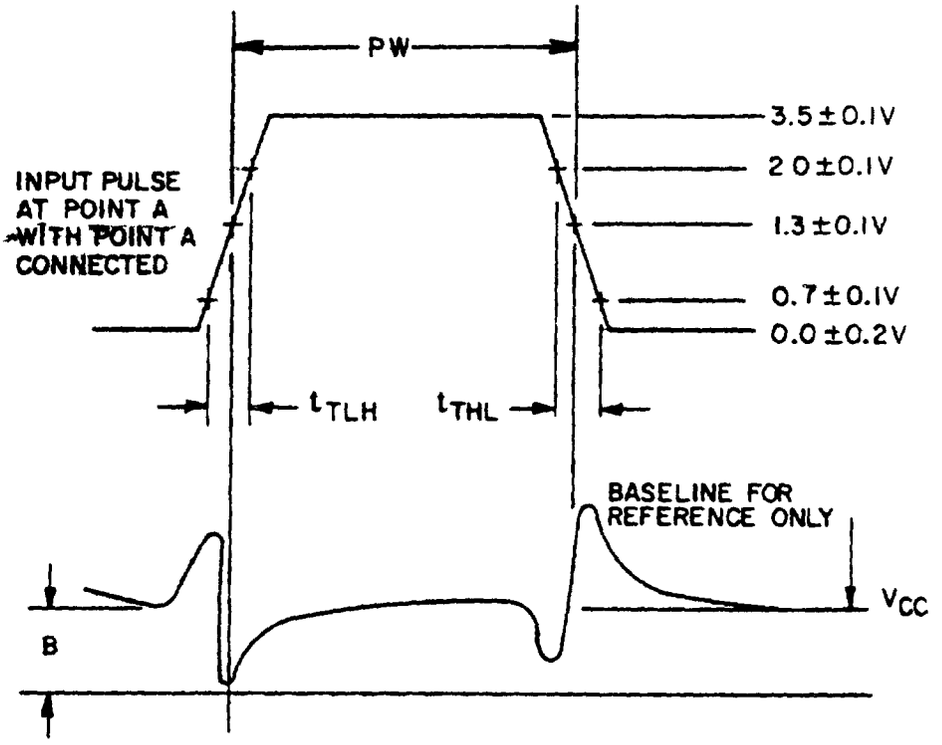
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NOTE: R1 = 50Ω ±5%, 1 W.

FIGURE 23. Power supply transient amplitude test circuit for table XVII.

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NOTES:
1. Input pulse parameters:
   PW = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
   PRR = 1 MHz ±10%.
2. B = Measurement from the baseline to the most negative excursion of the waveform.

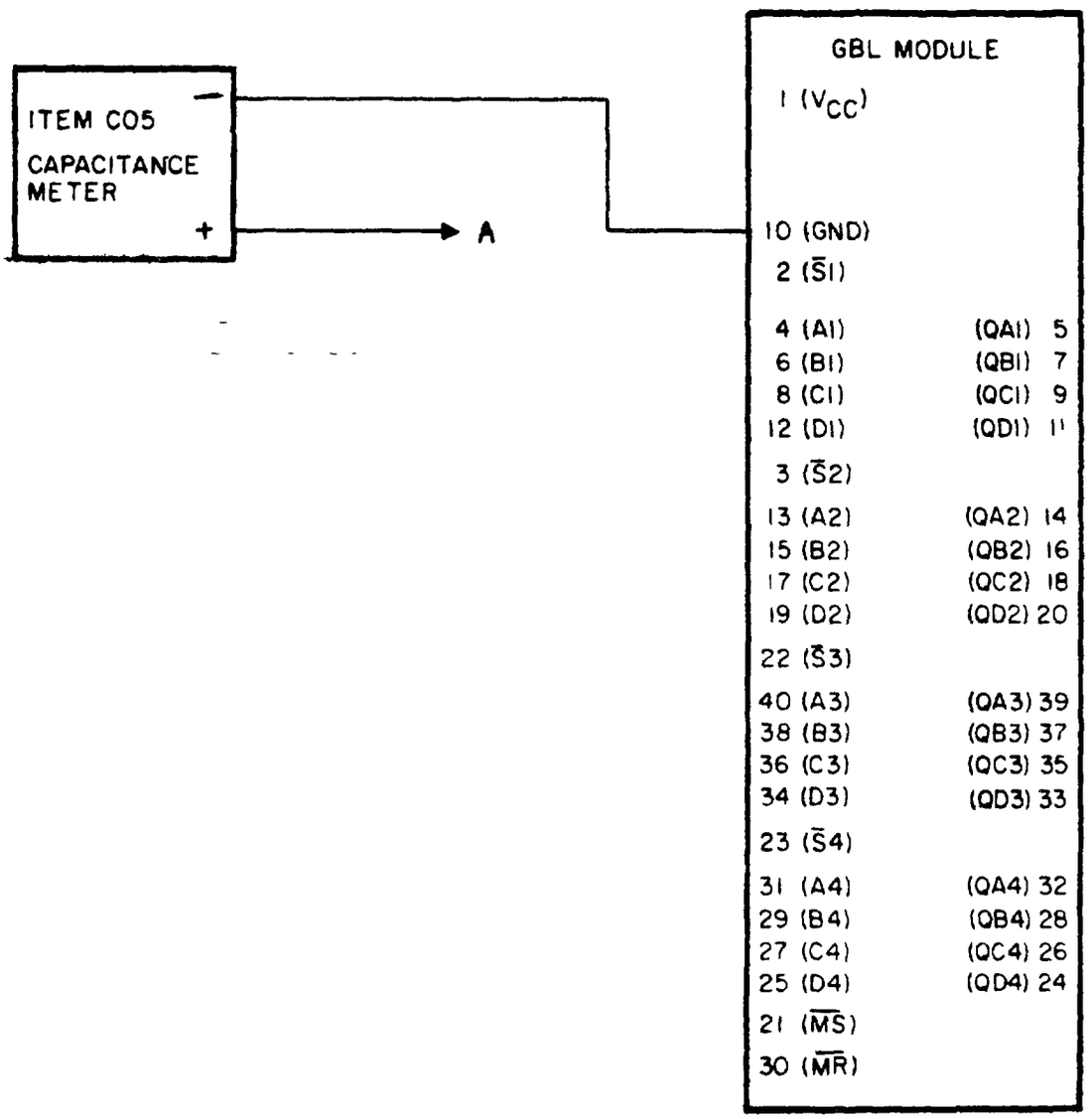
* FIGURE 24. Power supply transient amplitude test circuit waveforms for table XVII.

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TABLE XVIII. Input capacitance test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
PP1	A1 Input capacitance	4	2	15	pF	25
PP2	B1	6				
PP3	C1	8				
PP4	D1	12				
PP5	A2	13				
PP6	B2	15				
PP7	C2	17				
PP8	D2	19				
PP9	A3	40				
PP10	B3	38				
PP11	C3	36				
PP12	D3	34				
PP13	A4	31				
PP14	B4	29				
PP15	C4	27				
PP16	D4	25				
PP17	$\overline{S}1$	2				
PP18	$\overline{S}2$	3				
PP19	$\overline{S}3$	22				
PP20	$\overline{S}4$	23				
PP21	$\overline{WS}$	21	8	45		
PP22	$\overline{WR}$	30	8	45		

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 25. Input capacitance test circuit for table XVIII.

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